

Chapter 1

Multinational Financial Management: Opportunities and Challenges

The Multinational Enterprise (MNE) là doanh nghiệp có tổng kinh doanh tại hơn 1 quốc gia

- MNEs are firms - both for-profit and not-for-profit organizations - that have operations in more than one country and conduct their business through branches, foreign subsidiaries, joint ventures or affiliates with host country firms. VD: Vinfast bán xe ở Việt Nam, Mỹ, châu Âu & đầu tư nhà máy ở Mỹ; một kinh doanh ở có người
- Today's MNEs are dependent on emerging markets.

Financial Globalization and Risk mục tiêu của quản trị tài chính quốc tế

- The global financial marketplace is a combination of complex risks: (foreign exchange risks, movement of the exchange rate)
 - The international monetary system is an eclectic mix of floating and managed fixed exchange rates. China (renminbi) is more and more a dominant player. trong tài chính doanh nghiệp, forum comes do tỷ giá hối đoái ngoại, giữ các quốc gia khác nhau, luật chuyển tiền, sự ra đời của đồng tiền, chiến tranh, văn hóa, kế toán, thị trường
 - Monetary and fiscal policies are complicated by large fiscal deficits. lãi suất
 - Large and continuing balance of payments imbalances will shift the exchange rate landscape.
 - The dominant form of business may have shifted from the publicly-traded firm to the privately owned model.
 - Global capital markets are shrinking and changing.
 - Large inflows and outflows of capital travel through industrial and emerging markets complicating financial management.

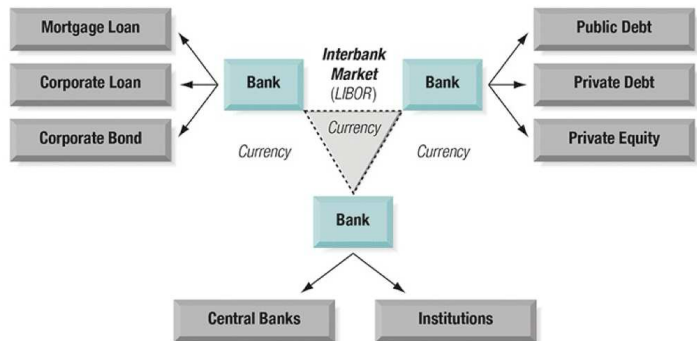
The Global Financial Marketplace

Assets, institutions, and linkages comprise one method to map global capital markets →

VD: Trái phiếu chính phủ Mỹ, cổ phiếu, khoản vay ngân hàng

- Assets** are debt securities issued by governments (e.g., U.S. Treasury Bonds). These form the baseline for other forms of financing.
- Institutions:** ngân hàng trung ương, ngân hàng thương mại, ngân hàng đầu tư
 - + central banks: create and control country's money supply
 - + commercial & investment banks: take deposits and extend loans to businesses
- Their health keeps the global financial system stable! VD: Mạng thanh toán quốc tế, thị trường ngoại tệ
- Linkages** are the interbank networks using currency. Without a ready exchange of currencies the market is hard-pressed to operate efficiently.

The global capital market is a collection of institutions (central banks, commercial banks, investment banks, not-for-profit financial institutions like the IMF and World Bank) and securities (bonds, mortgages, derivatives, loans, etc.), which are all linked via a global network—the *Interbank Market*. This interbank market, in which securities of all kinds are traded, is the critical pipeline system for the movement of capital.



The exchange of securities—the movement of capital in the global financial system—must all take place through a vehicle—currency. The exchange of currencies is itself the largest of the financial markets. The interbank market, which must pass-through and exchange securities using currencies, bases all of its pricing through the single most widely quoted interest rate in the world—LIBOR (the London Interbank Offered Rate).

The Market for Currencies

- Most currencies are quoted against the dollar as in “so many units per dollar”
- A few are quoted as “dollars per unit” due to custom e.g., \$/£ and \$/€.
- The price of any one country's currency in terms of another country's currency is called a foreign currency exchange rate.

là đồng tiền của 1 quốc gia nhưng được gọi tên bằng đồng tiền của quốc gia phát hành

Eurocurrencies and LIBOR

Eurocurrencies (a major linkage in the global and capital markets)

- These are domestic currencies of one country on deposit in a second country (anywhere outside the home country)
- The Eurocurrency markets serve 2 valuable purposes:
 - Eurocurrency deposits are an efficient and convenient money market device for holding excess corporate liquidity. (higher deposit rate offered in euro currency countries than the home country)
 - The Eurocurrency market is a major source of short-term bank loans to finance corporate working capital needs (including export and import financing)

Eurocurrency Interest Rates: LIBOR (lãi suất tham chiếu)

- In the Eurocurrency market, the reference rate of interest is the *London Interbank Offered Rate* (LIBOR) In any money market around the world Interest rate = LIBOR + 2%.
- This rate is the most widely accepted rate of interest used in standardized quotations, loan agreements, and financial derivatives transactions

currency exchange rate quotes: →

VD: - USD gửi tại sân gọi là Eurodollar
- USD gửi tại London cũng gọi là Eurodollar

March 23, 2015 Country	Currency	Symbol	Code	Currency Equal to 1 Dollar	Currency Equal to 1 Euro	Currency Equal to 1 Pound
Argentina	peso	Ps	ARS	8.7851	9.5990	13.1197
Australia	dollar	A\$	AUD	1.2744	1.3924	1.9032
Brazil	real	R\$	BRL	3.1610	3.4538	4.7206
Canada	dollar	C\$	CAD	1.2516	1.3676	1.8691
Chile	peso	\$	CLP	625.98	683.97	934.84
China	yuan	¥	CNY	6.2160	6.7919	9.2830
Czech Republic	koruna	Kc	CZK	25.0240	27.3424	37.3710
Denmark	krone	Dkr	DKK	6.8243	7.4666	10.1915
Egypt	pound	£	EGP	7.6301	8.3369	11.3948
Euro	euro	€	EUR	0.9152	1.0000	1.3668
India	rupee	Rs	INR	62.2650	68.0336	92.9870
Indonesia	rupiah	Rp	IDR	13,019.50	14,225.70	19,443.41
Israel	shekel	Shk	ILS	3.9655	4.3329	5.9221
Japan	yen	¥	JPY	119.765	130.861	178.858
Kenya	shilling	KSh	KES	91.90	100.41	137.24
Malaysia	ringgit	RM	MYR	3.6950	4.0373	5.5181
Mexico	new peso	\$	MXN	14.9583	16.3441	22.3387
New Zealand	dollar	NZ\$	NZD	1.3120	1.4335	1.9593
Nigeria	naira	₦	NGN	199.700	218.201	298.233
Norway	krone	Nkr	NOK	7.8845	8.6149	11.7747
Philippines	peso	₱	PHP	44.7900	48.9396	66.8897
Poland	zloty	—	PLN	3.7600	4.1084	5.6153
Russia	ruble	R	RUB	58.7450	64.1875	87.7302
Singapore	dollar	S\$	SGD	1.3671	1.4938	2.0416
South Africa	rand	R	ZAR	11.9100	13.0134	17.7865
South Korea	won	₩	KRW	1,114.65	1,217.92	1,664.63
Sweden	krona	SKr	SEK	8.5016	9.2892	12.6963
Switzerland	franc	Fr.	CHF	0.9669	1.0565	1.4440
Taiwan	dollar	T\$	TWD	31.4350	34.3473	46.9452
Thailand	baht	B	THB	32.5450	35.5601	48.6029
Turkey	lira	YTL	TRY	2.5477	2.7837	3.8048
United Kingdom	pound	£	GBP	0.6696	0.7316	1.0000
Ukraine	hryvnia	—	UAH	22.6500	24.7484	33.8257
Uruguay	peso	\$U	UYU	25.5300	27.8952	38.1267
United States	dollar	\$	USD	1.0000	1.0926	1.4934
Venezuela	Bolivar fuerte	Bs	VEB	6.2935	6.8766	9.3988
Vietnam	dong	đ	VND	21,525.00	23,519.23	32,145.66
Special Drawing Right	—	—	SDR	0.7199	0.7867	1.0752

Note that a number of different currencies use the same symbol (for example, both China and Japan have traditionally used the ¥ symbol, yen or yuan, meaning round or circle). All quotes are mid-rates, and are drawn from the *Financial Times*.

The Theory of Comparative Advantage

The theory of comparative advantage (expanded version of absolute advantage) provides a basis for explaining and justifying international trade in a model world assumed to enjoy:

- free trade; perfect competition; no uncertainty; costless information; and no government interference.

The theory contains the following features:

- Exporters in Country A sell G/S to unrelated importers in Country B
- Firms in Country A specialize in making products that can be produced relatively efficiently, given Country A's endowment of factors of production, that is, land, labor, capital, and technology
- Firms in Country B do likewise, given the factors of production found in Country B
- In this way the total combined output of A and B is maximized
- Because the factors of production cannot be moved freely from Country A to Country B, the benefits of specialization are realized through international trade
- The way the benefits of the extra production are shared depends on the terms of trade, the ratio at which quantities of the physical goods are traded
- Each country's share is determined by supply and demand in perfectly competitive markets in the two countries
- Neither Country A nor Country B is worse off than before trade, and typically both are better off, albeit perhaps unequally

For example: Việt Nam vs Japan có thể sản xuất gạo & máy móc:

Country	Gạo	Máy móc
VN	10 tấn	1 máy
Japan	6 tấn	3 máy

⇒ VN có comparative advantage về gạo: 10 tấn vs 6 tấn
 ⇒ Japan có comparative advantage về máy móc: 3 máy vs 1 máy
 comparative advantage khi 1 quốc gia 1 cân phải giảm nhất tuyệt đối; nước đó chỉ cần có opportunity cost thấp hơn hàng đối tượng 1 qđ

What is Different About International Financial Management?

Exhibit 1.3 summarizes the differences.

- Culture and history differ among countries
- Corporate governance
- Greater levels of foreign exchange and political risks
- Financial theory and applications are modified in the global versus domestic marketplace
- Specialized and complicated financial instruments become tools of the trade

Concept	International	Domestic
Culture, history, and institutions	Each foreign country is unique and not always understood by MNE management	Each country has a known base case
Corporate governance	Foreign countries' regulations and institutional practices are all uniquely different	Regulations and institutions are well known
Foreign exchange risk	MNEs face foreign exchange risks due to their subsidiaries, as well as import/export and foreign competitors	Foreign exchange risks from import/export and foreign competition (no subsidiaries)
Political risk	MNEs face political risk because of their foreign subsidiaries and high profile	Negligible political risks
Modification of domestic finance theories	MNEs must modify finance theories like capital budgeting and the cost of capital because of foreign complexities	Traditional financial theory applies
Modification of domestic financial instruments	MNEs utilize modified financial instruments such as options, forwards, swaps, and letters of credit	Limited use of financial instruments and derivatives because of few foreign exchange and political risks

Market Imperfections: A Rationale for the Existence of the Multinational Firm

- Market imperfection: movement of G/S is not allowed to move freely across the national border (tax,...) creating business opportunities for MNEs.

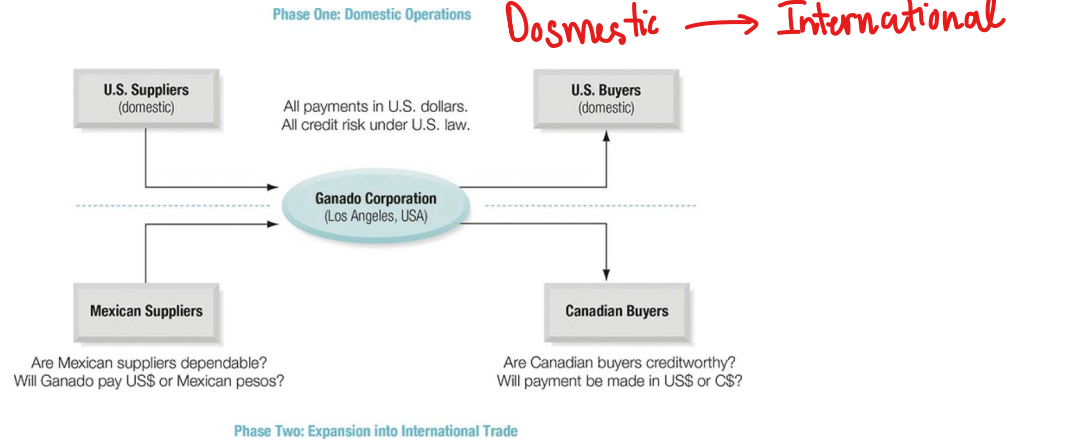
Why not opportunities for domestic firms? Cuz MNEs are superior compared to domestic firms. (win-win for MNEs and customers)

- MNEs strive to take advantage of imperfections in national markets for products, factors of production, and financial assets.
- Imperfections in the market for products translate into market opportunities for MNEs.
- Large international firms are better able to exploit such competitive factors as economies of scale, managerial and technological expertise, product differentiation, and financial strength than their local competitors.
- Strategic motives drive the decision to invest abroad and become a MNE and can be summarized under the following categories:
 - Market seekers (expand market shares)
 - Raw material seekers (enjoy cheaper product material, ex: Chinese firms)
 - Production efficiency seekers
 - Knowledge seekers (establish subsidiaries in other countries to learn sth from these countries)
 - Political safety seekers
- These categories are not mutually exclusive.

The Globalization Process

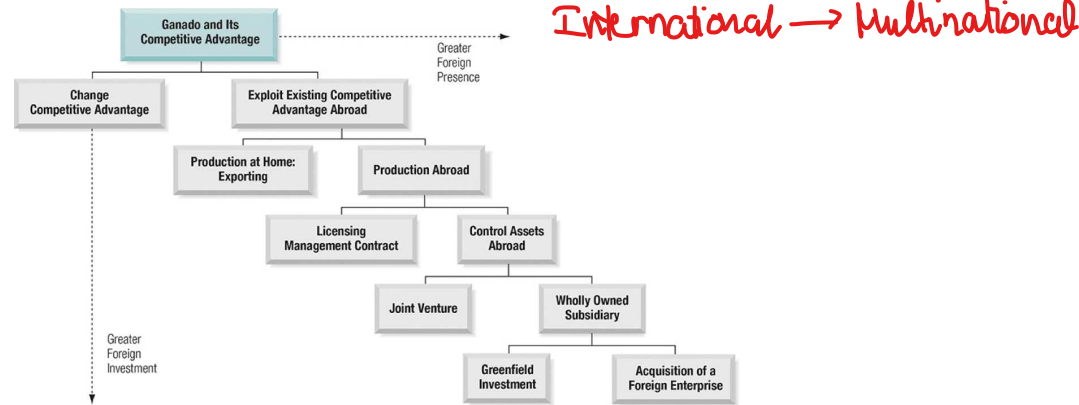
Stage I: early domestic phase growing into the international trade phase
(massan: do business with all domestic suppliers and consumers)

Exhibit 1.4 Ganado Corp: Initiation of the Globalization Process



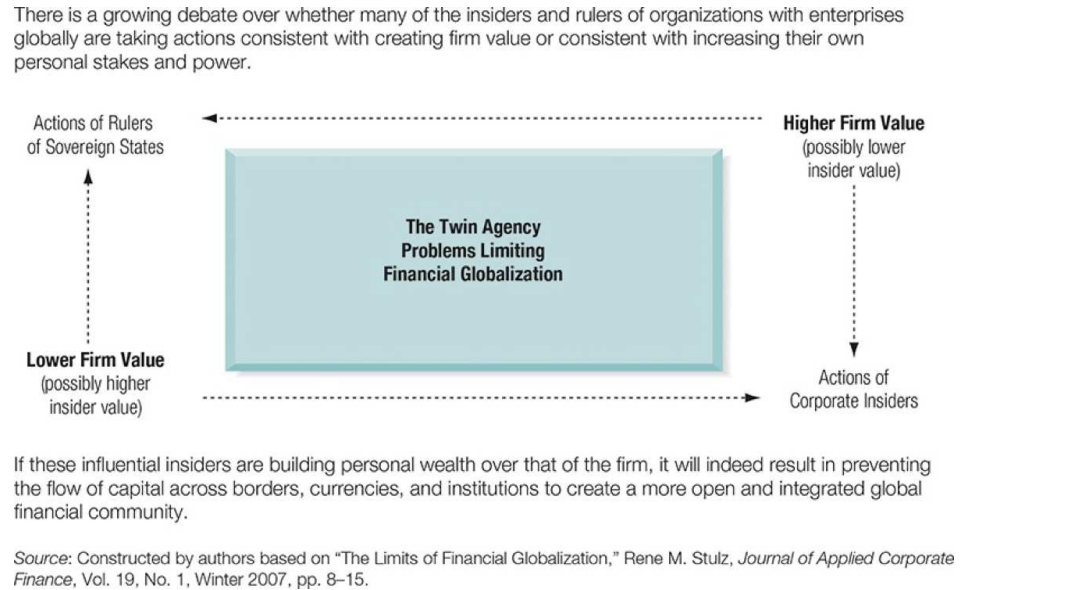
Stage II: A successful firm will continue to grow from simple international trade to the multinational phase characterized by production and investment both at home and abroad

Exhibit 1.5 Ganado's Foreign Direct Investment Sequence



Growth may be limited by the twin agency problems of corporate insiders and the rulers of sovereign states

Exhibit 1.6 Limits of Financial Globalization

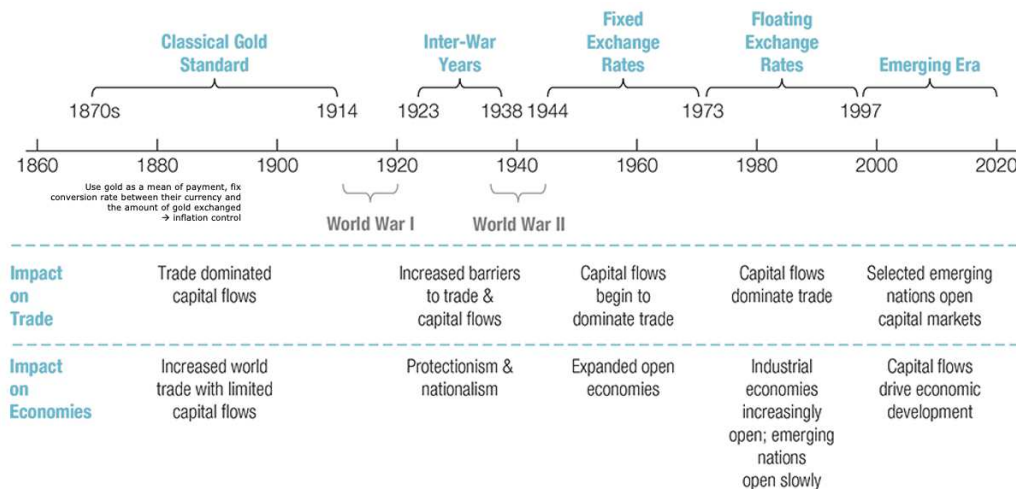


Chapter 2

The International Monetary System

History of the International Monetary System

The Evolution of the Global Monetary System



1. The Gold Standard (1876-1913)

- Gold has been a medium of exchange since 3000 BC
- "Rules of the game" were simple, each country set the rate at which its currency unit could be converted to a weight of gold
- Currency exchange rates were in effect "fixed"
- Expansionary monetary policy was limited to a government's supply of gold
- Was in effect until the outbreak of WWI when the free movement of gold was interrupted

2. The Inter-War Years & WWII (1914-1944)

- During this period, currencies were allowed to fluctuate over a fairly wide range in terms of gold and each other
- Increasing fluctuations in currency values became realized as speculators sold short weak currencies
- The U.S. adopted a modified gold standard in 1934
- During WWII and its chaotic aftermath the U.S. dollar was the only major trading currency that continued to be convertible

3. Bretton Woods and the International Monetary Fund (IMF) (1944)

- As WWII drew to a close, the Allied Powers met at Bretton Woods, New Hampshire to create a post-war international monetary system
- The Bretton Woods Agreement established a U.S. dollar based international monetary system and created two new institutions the International Monetary Fund (IMF) and the World Bank
- The International Monetary Fund is a key institution in the new international monetary system and was created to:
- Help countries defend their currencies against cyclical, seasonal, or random occurrences
- Assist countries having structural trade problems if they promise to take adequate steps to correct these problems
- Special Drawing Right (SDR) is the IMF reserve asset, currently a weighted average of four currencies
- The International Bank for Reconstruction and Development (World Bank) helped fund post-war reconstruction and has since then supported general economic development

4. Fixed Exchange Rates (1945-1973)

- The currency arrangement negotiated at Bretton Woods and monitored by the IMF worked fairly well during the post-WWII era of reconstruction and growth in world trade
- However, widely diverging monetary and fiscal policies, differential rates of inflation and various currency shocks resulted in the system's demise
- The U.S. dollar became the main reserve currency held by central banks, resulting in a consistent and growing balance of payments deficit which required a heavy capital outflow of dollars to finance these deficits and meet the growing demand for dollars from investors and businesses
- Eventually, the heavy overhang of dollars held by foreigners resulted in a lack of confidence in the ability of the U.S. to meet its commitment to convert dollars to gold
- The lack of confidence forced President Richard Nixon to suspend official purchases or sales of gold by the U.S. Treasury on August 15, 1971
- This resulted in subsequent devaluations of the dollar
- Most currencies were allowed to float to levels determined by market forces as of March 1973

5. The Floating Era (1973-1997)

- Since March 1973, exchange rates have become much more volatile and less predictable than they were during the "fixed" period
- There have been numerous, significant world currency events over the past 30 years
- The volatility of the U.S. dollar exchange rate index is illustrated in Exhibit 2.2

6. The Emerging Era (1997-present)

- Emerging market economics are multiplying in number and growing in complexity
- This results in a growing number of emerging market currencies

IMF Classification of Currency Regimes

Category 1: Hard Pegs

Countries that have given up their own sovereignty over monetary policy

- E.g., dollarization or currency boards

Category 2: Soft Pegs

AKA fixed exchange rates, with five subcategories of classification

Category 3: Floating Arrangements

Mostly market driven, these may be free floating or floating with occasional government intervention

Category 4: Residual

The remains of currency arrangements that don't well fit the previous categorizations

IMF Exchange Rate Classifications →

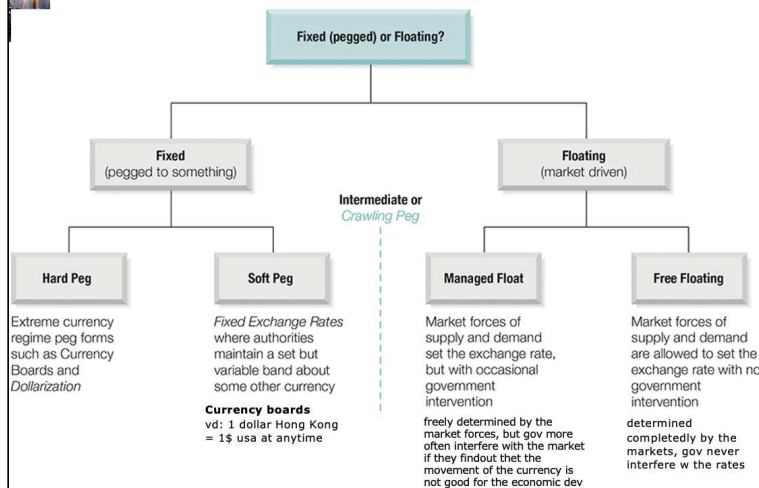
Rate Classification	2009 <i>de facto</i> System	Description and Requirements
Hard Pegs	Arrangement with no separate legal tender	The currency of another country circulates as the sole legal tender (formal dollarization), as well as members of a monetary or currency union in which the same legal tender is shared by the members.
	Currency board arrangement	A monetary arrangement based on an explicit legislative commitment to exchange domestic currency for a specific foreign currency at a fixed exchange rate, combined with restrictions on the issuing authority. Restrictions imply that domestic currency will be issued only against foreign exchange and that it remains fully backed by foreign assets.
Soft Pegs	Conventional pegged arrangement	A country formally pegs its currency at a fixed rate to another currency or a basket of currencies of major financial or trading partners. Country authorities stand ready to maintain the fixed parity through direct or indirect intervention. The exchange rate may vary $\pm 1\%$ around a central rate, or may vary no more than 2% for a six-month period.
	Stabilized arrangement	A spot market rate that remains within a margin of 2% for six months or more and is not floating. Margin stability can be met by either a single currency or basket of currencies (assuming statistical measurement). Exchange rate remains stable as a result of official action.
	Intermediate pegs: Crawling peg	Currency is adjusted in small amounts at a fixed rate or in response to changes in quantitative indicators (e.g., inflation differentials).
	Crawl-like arrangement	Exchange rate must remain with a narrow margin of 2% relative to a statistically defined trend for six months or more. Exchange rate cannot be considered floating. Minimum rate of change is greater than allowed under a stabilized arrangement.
Floating Arrangements	Pegged exchange rate within horizontal bands	The value of the currency is maintained within 1% of a fixed central rate, or the margin between the maximum and minimum value of the exchange rate exceeds 2%. This includes countries that are today members of the Exchange Rate Mechanism II (ERM II) system.
	Floating	Exchange rate is largely market determined without an ascertainable or predictable path. Market intervention may be direct or indirect, and serves to moderate the rate of change (but not targeting). Rate may exhibit more or less volatility.
	Free floating	A floating rate is freely floating if intervention occurs only exceptionally, and confirmation of intervention is limited to at most three instances in a six-month period, each lasting no more than three business days.
Residual	Other managed arrangements	This category is residual, and is used when the exchange rate does not meet the criteria for any other category. Arrangements characterized by frequent shifts in policies fall into this category.

Source: "Revised System for the Classification of Exchange Rate Arrangements," by Karl Habermeier, Anamaria Kokorenyne, Romain Veyrune, and Harald Anderson, IMF Working Paper WPI/09/211, International Monetary Fund, November 17, 2009.

Exhibit 2.4 shows **how these major regime categories translate in the global market.**

The vertical dashed line, the crawling peg, is the zone some currencies move into and out of depending on their relative currency stability.

Exhibit 2.4 Taxonomy of Exchange Rate Regimes

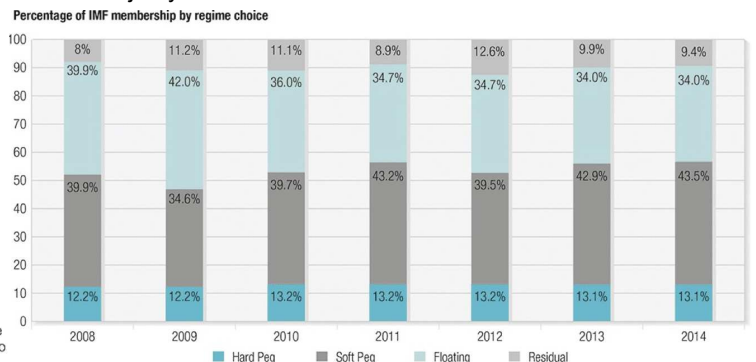


A Global Eclectic

Exhibit 2.5, the proportion of IMF member countries with floating regimes has been holding at about 34%.

Soft pegs continue to dominate.

Although the contemporary international monetary system is typically referred to as a "floating regime," it is clearly not the case for the majority of the world's nations.



Source: Data drawn from Annual Report on Exchange Arrangements and Exchange Restrictions 2014, International Monetary Fund, 2014, Table 3, Exchange Rate Arrangements 2008–2014.

Fixed Versus Flexible Exchange Rates

A nation's choice as to which currency regime to follow reflects national priorities about all facets of the economy, including:

- inflation,
- unemployment,
- interest rate levels,
- trade balances, and
- economic growth.

The choice between fixed and flexible rates may change over time as priorities change.

Countries would prefer a fixed rate regime for the following reasons:

- stability in international prices
- inherent anti-inflationary nature of fixed prices

However, a fixed rate regime has the following problems:

- Need for central banks to maintain large quantities of hard currencies and gold to defend the fixed rate
- Fixed rates can be maintained at rates that are inconsistent with economic fundamentals

Attributes of the “Ideal” Currency

Possesses three attributes, often referred to as the Impossible Trinity:

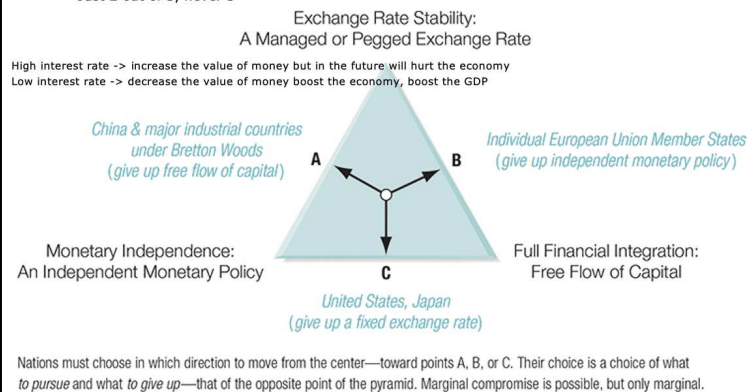
- Exchange rate stability
- Full financial integration
- Monetary independence

The forces of economics do not allow the simultaneous achievement of all three

Exhibit 2.6 illustrates how pursuit of one element of the trinity must result in giving up one of the other elements

Exhibit 2.6 The Impossible Trinity

Just 2 out of 3, never 3



A Single Currency for Europe: The Euro

- 12/ 1991, members of the European Union met at Maastricht, Netherlands, to finalize a treaty that changed Europe's currency future.
- This treaty set out a timetable and a plan to replace all individual ECU currencies with a single currency called the euro.
- To prepare for the EMU, a convergence criteria was laid out whereby each member country was responsible for managing the following to a specific level:
 - Nominal inflation rates
 - Long-term interest rates
 - Fiscal deficits
 - Government debt
- In addition, a strong central bank, called the European Central Bank (ECB), was established in Frankfurt, Germany.

Effects of the Euro

- The euro affects markets in three ways:
 - Cheaper transaction costs in the eurozone
 - Currency risks and costs related to uncertainty are reduced
 - All consumers and businesses both inside and outside the eurozone enjoy price transparency and increased price-based competition

Achieving Monetary Unification

- If the euro is to be successful, it must have a solid economic foundation.
- The primary driver of a currency's value is its ability to maintain its purchasing power.
- The single largest threat to maintaining purchasing power is inflation, so the job of the EU has been to prevent inflationary forces from undermining the euro.

The Greek/EU Debt Crisis

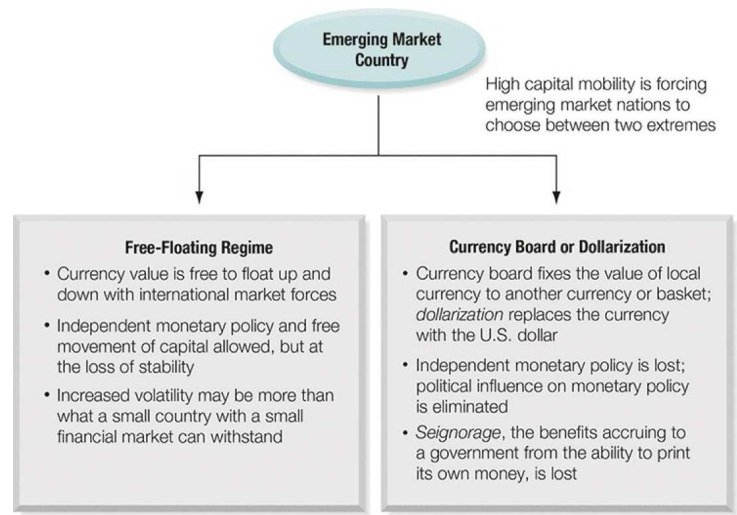
- The EU established exchange rate stability and financial integration with the adoption of the euro but each country gave up monetary independence.
- However, each country still controls its own fiscal policy and sovereign debt is denominated in euros and thus impacts the entire eurozone
- The ultimate outcome is still in question

Emerging Markets and Regime Choices

- A currency board (Hong Kong) exists when a country's central bank commits to back its monetary base – its money supply – entirely with foreign reserves at all times.
(always have to have enough US dollars reserver to back currency board policy)
Not allowed to bring out money.
- This means that a unit of domestic currency cannot be introduced (brought out) into the economy without an additional unit of foreign exchange reserves being obtained first.
→ before introduce any additional Hongkong dollars, they have to add more US dollar in the country foreign reserves
→ first fiscal balance (-)
control inflation (+)
 - Argentina moved from a managed exchange rate to a currency board in 1991
(huge deposit in fiscal balance trade give up)
 - In 2002, the country ended the currency board as a result of substantial economic and political turmoil
- Dollarization is the use of the U.S. dollar as the official currency of the country.
- One attraction of dollarization is that sound monetary and exchange-rate policies no longer depend on the intelligence and discipline of domestic policymakers.
 - Panama has used the dollar as its official currency since 1907
 - Ecuador replaced its domestic currency with the U.S. dollar in September 2000

Currency Regime Choices for Emerging Markets

- Some experts suggest countries will be forced to extremes when choosing currency regimes - either a hard peg or free-floating (Exhibit 2.8)
- 3 common features that make emerging market choices difficult:
 - weak fiscal, financial and monetary institutions
 - tendencies for commerce to allow currency substitution and the denomination of liabilities in dollars
 - the emerging market's vulnerability to sudden stoppages of outside capital flows



Globalizing the Chinese Renminbi

- The RMB's value has been carefully controlled but allowed to gradually revalue against the dollar (Exhibit 2.9)
- It is now quickly moving toward what most think is an inevitable role as a true international currency.
- The onshore market (RMB, CNY) is a two-tier market, with retail exchange and an interbank wholesale exchange.
 - Currency is traded through the China Foreign Exchange Trade System (CFETS), in which the People's Bank of China sets a daily central parity rate against the dollar (fixing).
 - Continues to be gradually deregulated.
- The offshore market for the RMB has grown out of a Hong Kong base (CNH, an unofficial symbol).
 - This offshore market has enjoyed preferred access to the onshore market by government regulators, both in acquiring funds and re-injecting funds (back-flow).
 - Growth in this market has been fueled by the issuance of RMB-denominated debt.

Exhibit 2.9 The Revaluation of the Chinese Yuan (1995-2015)

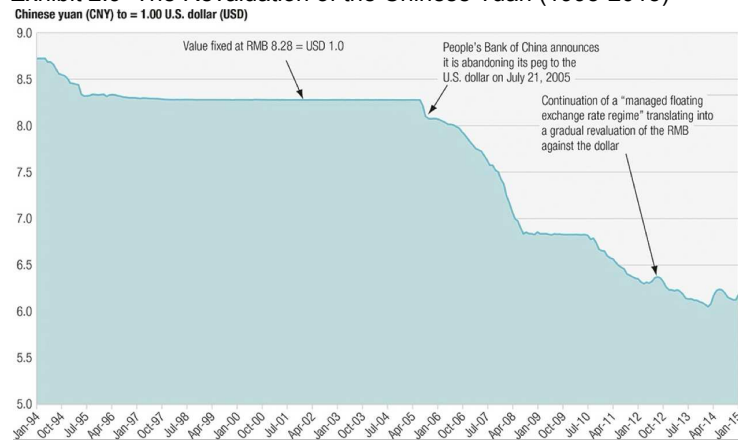
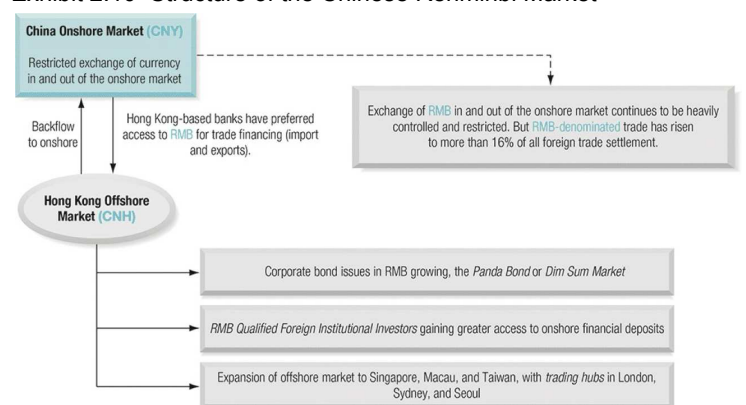


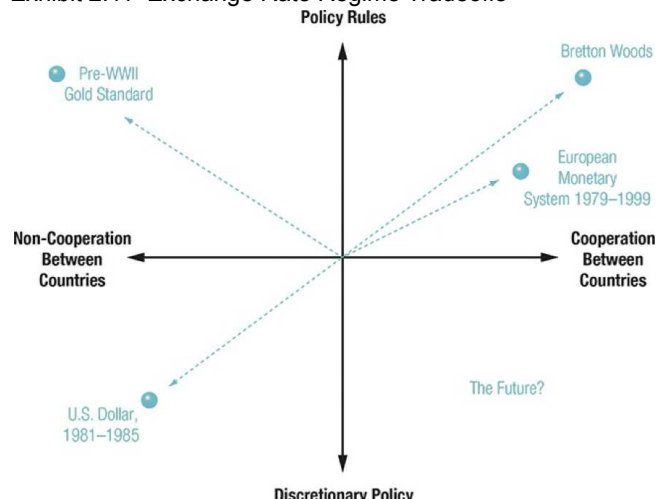
Exhibit 2.10 Structure of the Chinese Renminbi Market



Exchange Rate Regimes: What Lies Ahead?

- All exchange rate regimes must deal with the tradeoff between rules and discretion (vertical), as well as between cooperation and independence (horizontal) (see Exhibit 2.11)
- The pre WWI Gold Standard required adherence to rules and allowed independence
- The Bretton Woods agreement (and to a certain extent the EMS) also required adherence to rules in addition to cooperation
- The present system is characterized by no rules, with varying degrees of cooperation
- Many believe that a new international monetary system could succeed only if it combined cooperation among nations with individual discretion to pursue domestic social, economic, and financial goals

Exhibit 2.11 Exchange Rate Regime Tradeoffs



The Balance of Payments

The Balance of Payments

To sever natural interrelations is not to make oneself independent, but to isolate oneself completely. (Frederic Bastiat)

- The **measurement of all international economic transactions** between the residents of a country and foreign residents is called the balance of payments (BOP)
- BOP of Vietnam: transactions btw VN citizens and rest of the world
- BOP data is important for **government policymakers** and **MNEs** as it is a gauge of a nation's competitiveness or health (domestic and/or foreign) allow u to predict the future exchange rate btw 2 countries
- For a MNE, both home and host country BOP data is important as:
 - An indication of pressure on a country's foreign exchange rate
 - A signal of the imposition or removal of controls in various sorts of payments (dividends, interest, license fees, royalties and other cash disbursements)
 - A forecast of a country's market potential (especially in the short run)

Fundamentals of BOP Accounting

Statistical record of transaction between a country's citizens and the rest of the world; over a certain period of time (double entry bookkeeping)

- A BOP statement is a statement of cash flows over an interval of time.
- A **credit** is an event (the export of a G/S), records **foreign exchange earned** → **inflow of foreign exchange to the country**.
- A **debit** records **foreign exchange spent** (payments for imports or purchases of services) → **outflow of foreign exchange**.

Ex: u spent \$1,000 in Thailand debit record in BOP Vietnam

→ VNM exports milk to the US; VNM receives \$1,000k record as a credit on VN BOP.

- The BOP **must balance**. (credit = debit sum up all of the sub-account equal 0)
Foreign reserve account: after 1 year of operation, if Inflow > outflow foreign reserve account increase)
- It cannot be in disequilibrium unless something has not been counted or has been counted improperly.
- Therefore, it is incorrect to state that the BOP is in disequilibrium.

Balance	2005	2006	2007	2008	2009	2010	2011	2012	2013
Current Account Balance	-740	-807	-719	-687	-381	-444	-459	-461	-400
Capital Account Balance	13	-2	0	6	0	0	-1	7	0
Financial Account Balance	687	807	617	735	283	439	532	428	368
Net Errors and Omissions	26	-1	101	-50	150	7	-55	30	30
Reserves and Related	14	2	0	-5	-52	-2	-16	-4	3
Sum or Total	0	0	0	0	0	0	0	0	0

Source: International Monetary Fund, Balance of Payments Statistics Yearbook, 2013.

The Accounts of the BOP

The BOP has three major sub-accounts:

- the current account
- the capital account
- the financial account.

In addition, the **official reserves account** tracks government currency transactions. (in > out reserve increase)

A fifth account, the **net errors and omissions account** is produced to preserve the balance of the BOP. (transactions that not been counted)

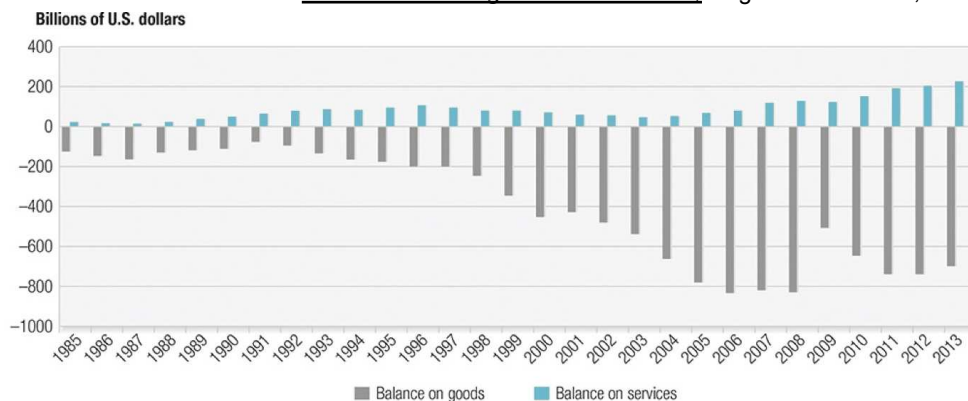
1. The Current Account

Includes **all international economic transactions with income or payment flows** occurring within **one year**, the current period. It consists of the following four subcategories:

- > **Goods trade** & import/ export of goods & merchandise
- > **Services trade**
- > **Income** (income of the whole country, include Vietnamese in & outside VN)
- > **Current transfers** (1 way transfer, remittances: VN citizens living in foreign countries but send \$ back to VN; financial aid...)

Current Account is typically dominated by the first component which is known as the Balance of Trade (BOT) even though it **excludes service trade**.

Exhibit 3.2 follows with U.S. **trade balance on goods and services**: Huge deficit in BOT; but surplus on BOS (fb,ig...)



Source: Balance of Payments Statistics Yearbook, International Monetary Fund, December 2014, p. 1,060.

- Merchandise trade is the original core of international trade.

- The manufacturing of goods was the basis of the industrial revolution and the focus of the theory of comparative advantage in international trade.
 - Declines in steel, automobiles, automotive parts, textiles, and shoe manufacturing have caused massive economic and social disruption
- ## 2. Capital account
- The capital and financial accounts of the balance of payments measure **all international economic transactions** of financial assets.
 - The capital account is made up of transfers of financial assets and the acquisition and disposal of non produced (land) /nonfinancial assets.
 - Has only been introduced recently as a separate account
- ## 3. Financial Account
- consists of 4 components
- **direct investment (FDI)**: degree of control is higher, (>10% of the shares)
 - **portfolio investment (FPI)**: lower degree of control (<10% of the shares)
 - **net financial derivatives**
 - **other asset investment.**
- Financial assets can be classified in different ways including:
 - the length of the life of the asset (maturity)
 - the nature of the ownership (public or private).
 - The financial account uses a degree of control over assets or operations to classify financial assets.
 - Direct Investment**
 - This is the net balance of capital dispersed from and into the U.S. for the purpose of exerting control over assets.
 - Source of concern over foreign investment in any country focuses on two topics: control and profit.
 - Some countries possess restrictions on what foreigners may own in their country.
 - Concerns over profit stem from the same argument.
 - Portfolio Investment**
 - This is the net balance of capital that flows in and out of the U.S. but does not reach the 10% threshold of direct investment.
 - The purchase of debt securities across borders is classified as portfolio investment because debt securities by definition do not provide the buyer with ownership or control.
 - Portfolio investment is motivated by a search for returns rather than to control or manage the investment.

Net Errors and Omissions Account ensures that the BOP actually balances.

Official Reserves Account is the total reserves held by official monetary authorities within the country.

- These reserves are normally composed of the major currencies used in international trade and financial transactions (hard currencies).
- The significance of official reserves depends generally on whether the country is operating under a fixed exchange rate regime or a floating exchange rate system.

Breaking the Rules: China's Twin Surpluses

- Most of the countries have opposite sides in financial & current account to ensure the balance in BOP
- China's highly unusual twin surplus in both the current and financial accounts (these relationships are typically inverse).
- Note that the financial account surplus fell sizably in 2012, only to rise to a record level in 2013—as a result of continuing deregulation of capital inflows into the country's economy.
- The reserves allow the Chinese government to manage the value of the Chinese yuan and its impact on Chinese competitiveness in the world economy.
- How big the China official reserve is\

BOP Impacts on Key Macroeconomic Rates

A country's balance of payments both impacts and is impacted by the three macroeconomic rates of international finance:

- exchange rates;
- interest rates;
- inflation rates

1. BOP and Exchange Rates:

Current Account Balance		Capital Account Balance		Financial Account Balance		Reserve Balance		Balance of Payments
$(X - M)$	+	$(CI - CO)$	+	$(FI - FO)$	+	FXB	=	BOP

X = exports of goods and services,

M = imports of goods and services,

CI = capital inflows,

CO = capital outflows,

FI = financial inflows,

FO = financial outflows,

FXB = official monetary reserves such as foreign exchange and gold.

- **Fixed Exchange Rate Countries:** government bears the responsibility to ensure that the BOP is near zero
- **Floating Exchange Rate Countries:** government has no responsibility to peg its foreign exchange rate
- **Managed Floats:** government take action to maintain their desired exchange rate values

2. The BOP and Interest Rates

- Relatively low real interest rates should normally stimulate an outflow of capital seeking higher rates elsewhere
- The opposite has occurred in the U.S. due to perceived growth opportunities and political stability - allowing it to finance its large fiscal deficit
- The favorable inflow on the financial account is diminishing while the current account balance is worsening - making the U.S. a bigger debtor nation vis-à-vis the rest of the world

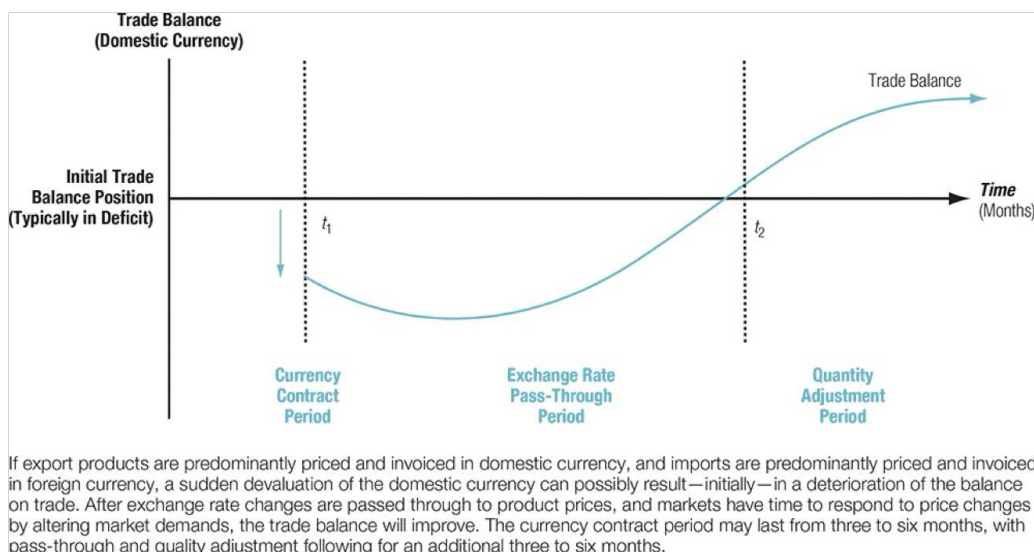
Handle inflation → higher the interest rate → investors move capital to VN to enjoy higher return in VN → surplus in Financial account → Supply of \$ go up, value of VND goes up against \$ → encourage import → deficit in BOT

3. The BOP and Inflation

- Imports have the potential to lower a country's inflation rate.
- Foreign competition substitutes for domestic competition to maintain a lower rate of inflation than might have been the case without imports.
- On the other hand, to the extent that lower-priced imports substitute for domestic production and employment, gross domestic product will be lower and the balance on the current account will be more negative.

Trade Balances and Exchange Rates

- A country's import and export of goods and services is affected by changes in exchange rates
- The transmission mechanism is in principle quite simple: changes in exchange rates change relative prices of imports and exports, and changing prices in turn result in changes in quantities demanded through the price elasticity of demand
- Theoretically, this is straightforward; in reality global business is more complex



Trade Adjustment to Exchange Rates: The J-Curve

Trade Balance Adjustment Path: The Equation

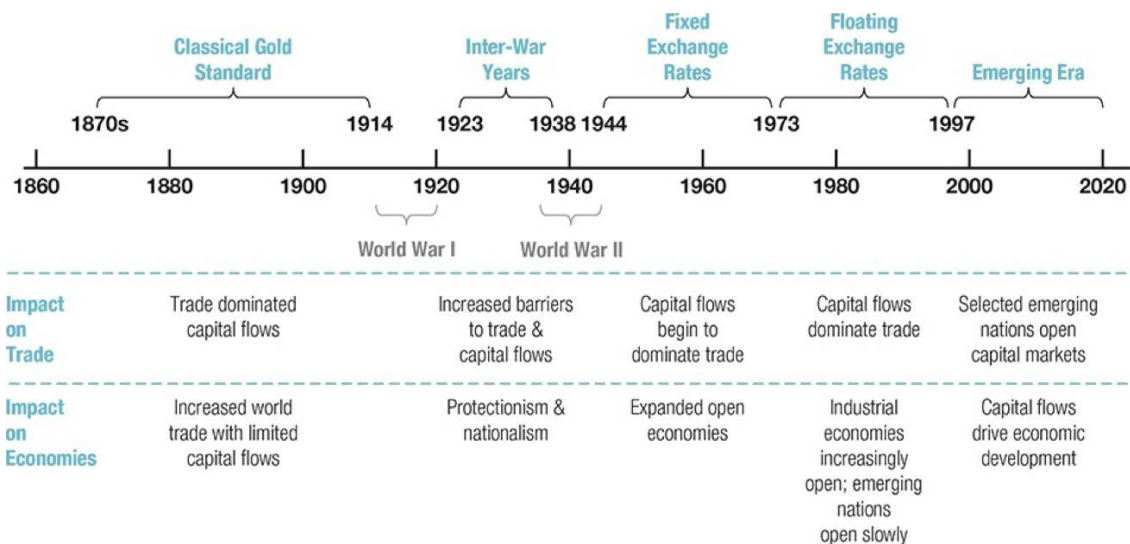
- A country's trade balance is essentially the net of import and export revenues.
- The U.S. trade balance, expressed in U.S. dollars, is then expressed as follows:

$$\text{U.S. Trade Balance} = P_x^{\$} Q_x - (S^{\$/fc} P_M^{fc} Q_M) \quad \text{fc: foreign currency}$$

Capital Mobility

- The degree to which **capital moves freely across borders** is critically important to a country's balance of payments BOP

boost country's economic dev (encourage investors to invest) → rapid inflow of capital	capital mobility increases → more susceptible to a banking crisis.
--	--
- The United States' financial account surplus has at least partially offset the current account deficits over the last 20 or more years*
- China has run a surplus in each of these accounts in recent years*
- The free flow of capital in and out of an economy can **potentially destabilize economic activity** or can **contribute significantly to an economy's development**
- Thus, Bretton Woods Agreement was careful to promote free movement of capital for current account transactions (e.g., foreign exchange or deposits) but less so for capital account transactions (e.g., foreign direct investment)
- 1970s-1990s saw growth in capital openness, the financial crisis of 1997/1998 stopped that due to destructive capital outflows and contagion
- The authors argue that the post-1860 era can be subdivided into four distinct periods with regard to capital mobility.
 - 1860-1914:** continuously increasing capital mobility as gold standard was adopted & international trade relations were expanded
 - 1914-1945:** global economic destruction, isolationist economic policies, negative effect on capital movement between countries
 - 1945-1971:** Bretton Woods era saw a great expansion of international trade
 - 1971-2097:** floating exchange rates, economic volatility, rapidly expanding cross-border capital flows



Capital Controls

A **capital control** is any restriction that limits or alters the rate or direction of capital movement into or out of a country

Free movement of capital is more the exception than the rule

Dutch Disease is the name given to the problem of a substantial currency appreciation due to the demand for a specific natural resource faced by several resource-rich smaller nations

VN: into is ok but not allow outflow

Several methods of and purposes for capital controls →

Control Purpose	Method	Capital Flow Controlled	Example
General Revenue/ Finance War Effort	Controls on capital outflows permit a country to run higher inflation with a given fixed-exchange rate and also hold down domestic interest rates.	Outflows	Most belligerents in WWI and WWII
Financial Repression/ Credit Allocation	Governments that use the financial system to reward favored industries or to raise revenue, may use capital controls to prevent capital from going abroad to seek higher returns.	Outflows	Common in developing countries
Correct a Balance of Payments Deficit	Controls on outflows reduce demand for foreign assets without contractionary monetary policy or devaluation. This allows a higher rate of inflation than otherwise would be possible.	Outflows	U.S. interest equalization tax 1963–1974
Correct a Balance of Payments Surplus	Controls on inflows reduce foreign demand for domestic assets without expansionary monetary policy or revaluation. This allows a lower rate of inflation than would otherwise be possible.	Inflows	German Bardepot Scheme 1972–1974
Prevent Potentially Volatile Inflows	Restricting inflows enhances macroeconomic stability by reducing the pool of capital that can leave a country during a crisis.	Inflows	Chilean <i>encaje</i> 1991–1998
Prevent Financial Destabilization	Capital controls can restrict or change the composition of international capital flows that can exacerbate distorted incentives in the domestic financial system.	Inflows	Chilean <i>encaje</i> 1991–1998
Prevent Real Appreciation	Restricting inflows prevents the necessity of monetary expansion and greater domestic inflation that would cause a real appreciation of the currency.	Inflows	Chilean <i>encaje</i> 1991–1998
Restrict Foreign Ownership of Domestic Assets	Foreign ownership of certain domestic assets—especially natural resources—can generate resentment.	Inflows	Article 27 of the Mexican Constitution
Preserve Savings for Domestic Use	The benefits of investing in the domestic economy may not fully accrue to savers so the economy as a whole can be made better off by restricting the outflow of capital.	Outflows	—
Protect Domestic Financial Firms	Controls that temporarily segregate domestic financial sectors from the rest of the world may permit domestic firms to attain economies of scale to compete in world markets.	Inflows and Outflows	—

Source: "An Introduction to Capital Controls," Christopher J. Neely, *Federal Reserve Bank of St. Louis Review*, November/December 1999, p. 16.

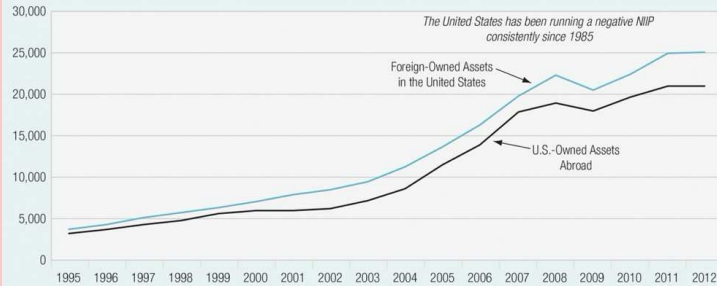
Capital Flight

- Capital flight: the rapid outflow of capital in opposition to or in fear of domestic political and economic conditions and policies, is one of the problems that capital controls are designed to control.
- Although it is not limited to heavily indebted countries, the rapid and sometimes illegal transfer of convertible currencies out of a country poses significant economic and political problems.
- Many heavily indebted countries have suffered significant capital flight.

Globalization of Capital Flows: Capital inflows are short-term in duration; Even mature markets can encounter crisis

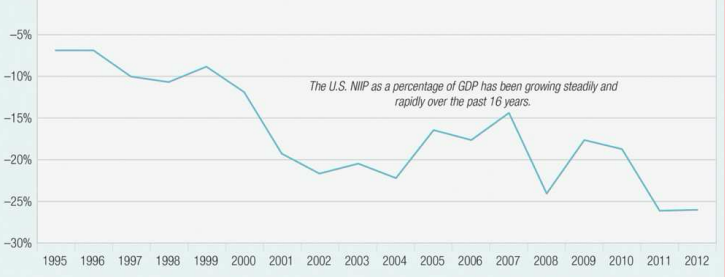
U.S. Net International Investment Position (NIIP)

Billions of U.S. dollars



U.S. NIIP as Percentage of Gross Domestic Product

0%



The Foreign Exchange Market

Functions of the Foreign Exchange Market

The foreign exchange market is the mechanism by which participants:

- transfer purchasing power between countries;
 - obtain or provide credit for international trade transactions; and
 - minimize exposure to the risks of exchange rate changes.
- foreign exchange market provides “hedging” facilities for transferring foreign exchange risk to someone else who is more willing to carry risk.

3 functions of money: as a unit of account, as a store of value, and as a medium of exchange.

Structure of the Foreign Exchange Market

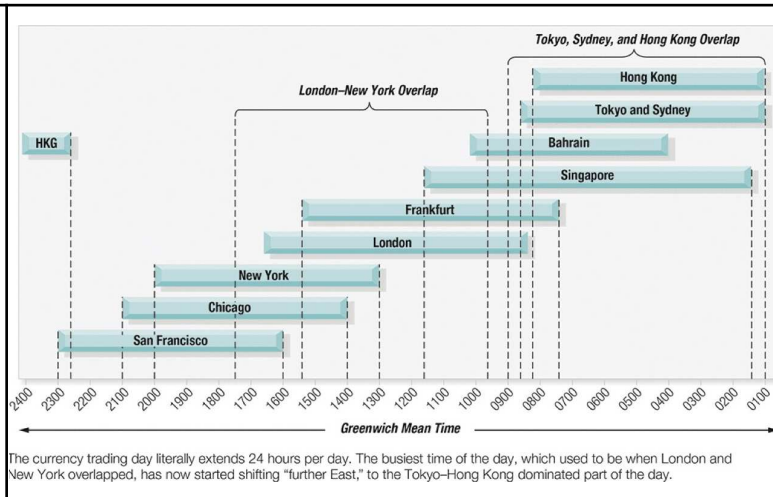
Global foreign exchange market today is rapidly changing due to:

- electronic trading platforms;
- algorithmic trading programs and routines
- increasing role of currencies as an asset class.

When the Asian-based trading centers overlap, the global currency markets exhibit the greatest depth and liquidity.

Global trading is a 24-hour-a-day process as shown in →
The world's trading day starts each morning in Sydney and Tokyo; and ends in San Francisco and Los Angeles.

proprietary trading: large international banks operate foreign exchange trading rooms in each major geographic trading center in order to serve both their customers and themselves



Trading Platforms and Processes

Even with continuous trading, a “closing price” is needed for a variety of record keeping and contractual purposes. Automated trading is growing, although human interaction is still used for some of the largest currency transactions.

Market Participants

Shift from a two-tier market (the interbank or wholesale market, and the client or retail market) to a single-tier market.

Participants in the foreign exchange market can be simplistically divided into 2 major groups:

- Those trading currency for commercial purposes, liquidity seekers (wish to secure currency for transactions)
- Those trading for profit, profit seekers. (much better informed about the market)
→ profit seekers generally profit from the liquidity seekers.

5 broad categories of institutional participants operate in the market:

1. Bank and Nonbank Foreign Exchange Dealers
 - Profit from buying foreign exchange at a bid price and reselling it at a slightly higher ask/offer price.
→ Competition among dealers worldwide narrows the spread between bids and offers
→ contributes to making the foreign exchange market “efficient”
 - Dealers in the foreign exchange department of large international banks often function as “market makers.”
 - These dealers stand willing at all times to buy and sell those currencies in which they specialize and thus maintain an “inventory” position in those currencies. → taking risk
 - Small- to medium-size banks and institutions are likely to participate but not to be market makers in the interbank market.
→ They often buy from and sell to larger institutions → offset retail transactions with their own customers or to seek short-term profits for their own accounts.
2. Individuals and Firms
 - Individuals (tourists,...) and firms (importers, exporters and MNEs) conduct commercial and investment transactions in the foreign exchange market.
 - Their use of the foreign exchange market is necessary but nevertheless incidental to their underlying commercial or investment purpose.
 - Some of the participants *use the market* to “*hedge*” foreign exchange risk.
3. Speculators and Arbitraders
 - *Speculators and arbitrageurs* seek to profit from trading in the market itself.
 - They *operate in their own interest*, no need/ obligation to serve clients or ensure a continuous market.
 - **dealers** seek the bid/ask spread
 - **speculators** seek all the profit from exchange rate changes
 - **arbitrageurs** try to profit from simultaneous exchange rate differences in different markets.

Traders employed by those banks conduct a large proportion of speculation and arbitrage on behalf of major banks.

→ banks act both as exchange dealers and as speculators and arbitrageurs.

4. Central Banks and Treasuries

- Use the market to acquire or spend their country's foreign exchange reserves as well as to influence the price at which their own currency is traded. → *foreign exchange intervention*.
- Act to support the value of their own currency because of policies adopted at the national level or because of commitments entered into through membership in joint agreements such as the European Monetary System.
- The motive is not to earn a profit as such, but rather to influence the foreign exchange value of their currency in a manner that will benefit the interests of their citizens.
- As willing loss takers, central banks and treasuries differ in motive from all other market participants.

→ Not to earn profit but to maintain/manipulate the exchange rate

5. Foreign Exchange Brokers

- Foreign exchange **brokers** are agents who *facilitate trading* between dealers without themselves becoming principals in the transaction → charge a small commission for this service.
→ brokers do not take risk (*do not do business for themselves, but to match potential seller w buyers to earn commission*)
- Dealers use brokers to expedite the transaction and to remain anonymous, since the identity of participants may influence short-term quotes.

6. Continuous Linked Settlement

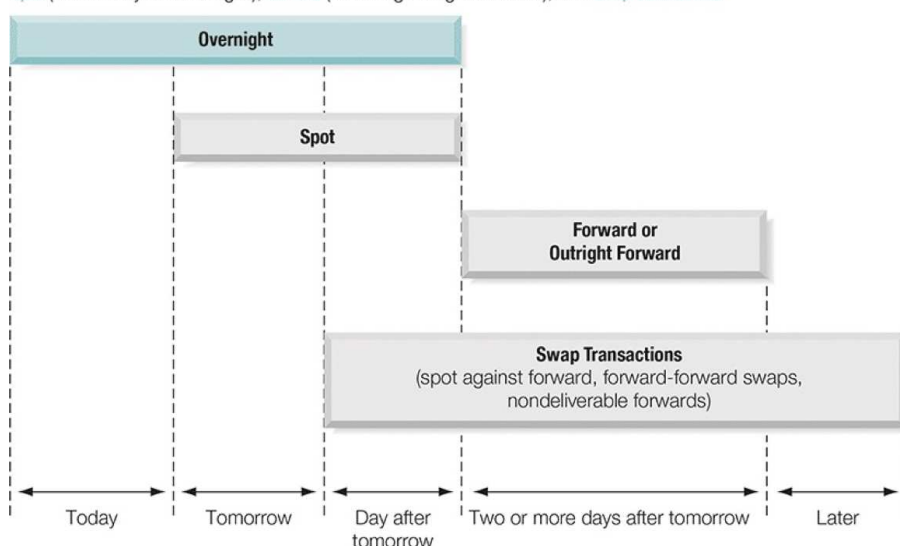
- In 2002, CLS system was introduced.
- CLS eliminates losses if either party of a foreign exchange transaction is unable to settle with the other party.
- It links the number of settlement systems, which operate on a real-time basis, and is expected to eventually result in same-day settlement, replacing the traditional two-day transaction period.

Transactions in the Foreign Exchange Market

1. Spot transaction: purchase of foreign exchange with delivery and payment between banks taking place **on the second following business day**.

- The date of settlement (payment day/ value day) is referred to as the value date. The day buyers make payment n sellers receive payment.
- On the value date, most dollar transactions in the world are settled through the computerized Clearing House Interbank Payments System (CHIPS) in NY, which calculates net balances owed by any one bank to another and which facilitates payment of those balances by 6:00 p.m. The price is determined today, but differs in the day payment is made.

Foreign exchange operations are defined by the timing—the *future date*—set for delivery. There are in principle three major categories of over-the-counter transactions categorized by future delivery: *spot* (which may be overnight), *forward* (including outright forward), and *swap transactions*.



2. Forward transaction

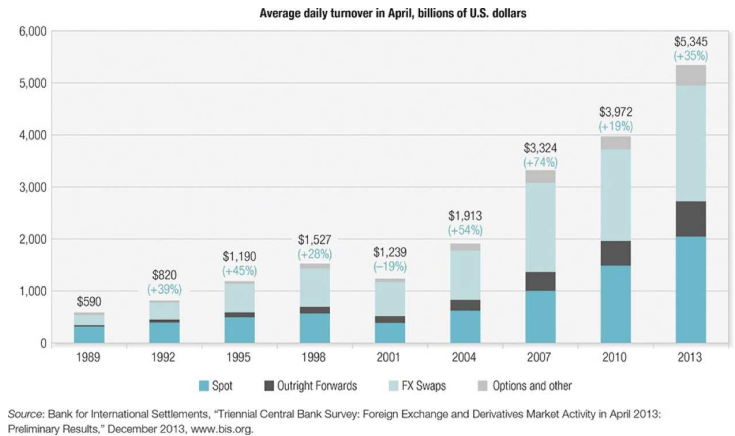
- An outright forward transaction (forward) requires delivery **at a future** value date of a specified amount of one currency for a specified amount of another currency (*price made today*)
- The exchange rate is established **at the time of the agreement**, but **payment and delivery are not required until maturity** (in the future)
- Forward exchange rates are usually quoted for value dates of 1, 2, 3, 6, 12 months.
- Buying forward and selling forward describe the **same transaction** (the difference is the order in which currencies are referenced.)

3. Swap transaction:

- The simultaneous purchase and sale of a given amount of foreign exchange for two different value dates.
 - Both purchase and sale are conducted with the same counterparty.
 - Some different types of swaps are:
 - Spot against forward: (most common): hợp đồng kết hợp giữa một hợp đồng giao ngay và một hợp đồng kỳ hạn
Dealer buys a currency in the spot market (at the spot rate) and simultaneously sells the same amount back to the same bank in the forward market (at the forward exchange rate).
→ a single transaction with just one counterparty → dealer incurs no unexpected foreign exchange risk
 - Forward-forward: là hợp đồng kết hợp giữa hai hợp đồng kỳ hạn được thực hiện cùng một lúc tuy nhiên thời gian đáo hạn của hai hợp đồng này khác nhau.
 - Non deliverable forwards (NDF): Hợp đồng kì hạn không giao dịch là một hợp đồng kì hạn bằng tiền mặt thay vì giao nhận tài sản gốc và thường là ngắn hạn.
→ Is a common derivative offered by the largest providers of foreign exchange derivatives. NDFs possess the same characteristics and documentation requirements as traditional forward contracts, except that they are settled only in U.S. dollars. NDFs are traded and settled outside the country of the subject currency → beyond the control of the government.
→ Problems: "fixing of spot rate on the fixing date," the spot rate at the end of the contract used to calculate the settlement. In times of financial crisis, the government of the subject currency may suspend foreign exchange trading in the spot market for an extended period. Without an official fixing rate, the NDF cannot be settled.

Size of the Foreign Exchange Market

- 2013: a survey conducted by the Bank for International Settlements (BIS) in conjunction with central banks around the world estimated the daily global net turnover in the foreign exchange market to be \$5.3 trillion.
- The primary driver of rapid foreign exchange growth in recent years is attributed to increasing profit seeker activity facilitated by electronic trading and access to the greater market.
- Exhibit 5.3 shows an average annual growth of 9.6% per year since 1989.



Geographical Distribution

- The United Kingdom (London) continues to be the world's major foreign exchange market in traditional foreign exchange market activity with 40.9% of the global market.

Currency Composition

- The "Big three" (dollar, euro, yen) continue to dominate global currency trades, totaling roughly 92% of all trading surveyed. Today: big 5 (dollar, euro, yen, british pound, chinese remedy)
- Emerging market currencies (Mexican peso, Chinese renminbi, Russian ruble) could soon change the dynamics of global currency trading.

Foreign Exchange Rates and Quotations

- A foreign exchange rate is the price of one currency expressed in terms of another currency.
- A foreign exchange quotation (or quote) is a statement of willingness to buy or sell at an announced rate.
- Quotations may be designated by *traditional currency symbols* or by *ISO codes*:

Currency	Traditional Symbol	ISO 4217 Code
U.S. dollar	\$	USD
European euro	€	EUR
Great Britain pound	£	GBP
Japanese yen	¥	JPY
Mexican peso	Ps	MXN

All electronic trading between institutions in the global marketplace uses the three-letter ISO codes. The paper currency of most countries continues to be represented using the country's traditional currency symbol.

- Every currency exchange involves two currencies: the *base* or *unit currency* (CUR1), and the *price* or *quote currency* (CUR2):
CUR1 / CUR2
 - Currency to the left of the slash: base currency/ unit currency, always a single unit
 - Currency to the right of the slash: price currency/ quote currency.
 - 1: base/unit currency, 2: price/quote currency (if the number go first → inverse)
- The quotation indicates the number of units of CUR2 required in exchange for receiving one unit of CUR1.
- EUR / USD 1.2174** → 1 *eur* = 1.2175 *usd*
 - designates the euro (EUR) as the *base currency*, the dollar (USD) as the *price currency*.
- The exchange rate is **USD 1.2174 = EUR 1.00**.
 - eur: base/ unit currency; dollar: price/quote currency
- VND 24,000/ USD → usd: unit currency

•Exhibit 5.6 provides an overview of the multitude of terms used to quote currencies.

European terms

Foreign currency price of one dollar (USD)

USD/EUR 0.8214
or
USD 1.00 = EUR 0.8214

American terms

U.S. dollar price of one euro (EUR)

EUR/USD 1.2174
or
EUR 1.00 = USD 1.2174

USD is the base or unit currency
EUR is the quote or price currency

EUR is the base or unit currency
USD is the quote or price currency

$$\frac{1}{\text{EUR } 0.8214 / \text{USD}} = \text{USD } 1.2174 / \text{EUR}$$

European terms(VN) the quoting of the quantity of a specific currency per one U.S. dollar, is *most common*.

American terms: 2 major exceptions: the euro and the U.K. pound sterling:
→ the U.S. dollar price of one euro; the U.S. dollar price of one pound sterling.

- American terms are also utilized in quoting rates for most foreign currency options and futures, as well as in retail markets that deal with tourists.

Direct and Indirect Quotations

In this pair of definitions, the home or base country of the currencies being discussed is critical.

- **Direct quote:** *home currency price of a unit of foreign currency.* (VND 24,000/ USD).
→ the price of a foreign currency in domestic currency units.
- **Indirect quote:** *foreign currency price of a unit of home currency.*
→ the price of the domestic currency in foreign currency units.

The form of the quote depends on what the speaker regards as "home"

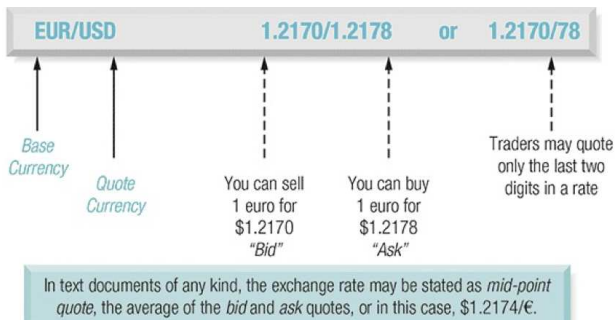
in Paris: EUR 0.8214 = USD 1.00 → direct quote on the dollar or a price quote on the dollar.
In France: home currency is the euro (the price); foreign currency is the dollar (the unit),
"0.8214 euros per dollar," or "it will cost me 0.8214 euros to get one dollar." → European terms.
in U.S: USD 1.2174 = EUR 1.00 → home currency is the dollar (the price); foreign currency is the euro (the unit),
→ direct quote on the euro (the home currency price of one unit of foreign currency)
→ indirect quote on the dollar (the foreign currency price of one unit of home currency).
- "I will pay \$1.2174 dollars per euro." → American terms.

Bid and Ask Rates. Interbank quotations are given as a bid and ask.

- A **bid** is the price (i.e., exchange rate) in one currency at which a **dealer is willing to buy** another currency.
- An **ask** is the price (i.e., exchange rate) at which a **dealer is willing to sell** the other currency.

Dealers bid (buy) at one price and ask (sell) at a slightly higher price, making their profit from the spread between the buying and selling prices. The bid-ask spread may be quite large for currencies that are traded infrequently, in small volumes, or both.

A bid for one currency is also the offer for the opposite currency. .



For example, the *Wall Street Journal* would quote the following currencies as follows:

	Last Bid		Last Bid
Euro (EUR/USD)	1.2170	Brazilian Real (USD/BRL)	1.6827
Japanese Yen (USD/JPY)	83.16	Canadian Dollar (USD/CAD)	0.9930
U.K. Pound (GBP/USD)	1.5552	Mexican Peso (USD/MXN)	12.2365

Intermarket Arbitrage

Cross rates:

Many currency pairs are only *inactively traded* (VND, can't buy in the global market, NDF), so their exchange rate is determined through their relationship to a widely traded third currency

Cross rates can be used to check on opportunities for intermarket arbitrage. Suppose the following exchange rates are quoted:.

February 18, 2015					
Country	Currency	Symbol	Code	USD equivalent	Currency per USD
Americas					
Argentina	peso	Ps	ARS	0.1151	8.6904
Brazil	real	R\$	BRL	0.3529	2.8338
Canada	dollar	C\$	CAD	0.8029	1.2455
Chile	peso	\$	CLP	0.001614	619.6
Colombia	peso	Col\$	COP	0.0004114	2430.74
Ecuador	U.S. dollar	\$	USD	1	1
Mexico	new peso	\$	MXN	0.0673	14.8638
Peru	new sol	S/.	PEN	0.3241	3.0855
Uruguay	peso	\$U	UYU	0.04062	24.62
Venezuela	bolivar fuerte	Bs	VND	0.15885497	6.2951
Asia-Pacific					
Australia	dollar	A\$	AUD	0.7812	1.2801
China	yuan	¥	CNY	0.1598	6.2564
Hong Kong	dollar	HK\$	HKG	0.1289	7.7586
India	rupee	₹	INR	0.01611	62.0568
Indonesia	rupiah	Rp	IDR	0.0000778	12857
Japan	yen	¥	JPY	0.00842	118.79
Malaysia	ringgit	RM	MYR	0.2764	3.6175
New Zealand	dollar	NZ\$	NZD	0.7545	1.3254
Pakistan	rupee	Rs.	PKR	0.00986	101.45
Philippines	peso	₱	PHP	0.0226	44.2375
Singapore	dollar	S\$	SGD	0.7376	1.3558
South Korea	won	₩	KRW	0.000902	1108.62
Taiwan	dollar	T\$	TWD	0.03159	31.65
Thailand	baht	฿	THB	0.0307	32.57
Vietnam	dong	đ	VND	0.00004688	21333
Europe					
Czech Republic	koruna	Kc	CZK	0.04182	23.912
Denmark	krona	Dkr	DKK	0.1531	6.5326
Euro area	euro	€	EUR	1.1398	0.8774
Hungary	forint	Ft	HUF	0.00372537	268.43
Iceland	krona	kr	ISK	0.007569	132.11
Norway	krona	Nkr	NOK	0.1328	7.5279
Poland	zloty	—	PLN	0.2724	3.671
Romania	leu	L	RON	0.2562	3.9039
Russia	ruble	₽	RUB	0.01624	61.579
Sweden	krona	SKr	SEK	0.1196	8.3619
Switzerland	franc	Fr.	CHF	1.061	0.9425
Turkey	lira	₺	TRY	0.4093	2.4433
United Kingdom	pound	£	GBP	1.5438	0.6478
Middle East/Africa					
Bahrain	dinar	—	BHD	2.6524	0.377
Egypt	pound	£	EGP	0.1312	7.6241
Israel	shekel	Shk	ILS	0.2595	3.8542
Kuwait	dinar	—	KWD	3.3835	0.2956
Saudi Arabia	riyal	SR	SAR	0.2665	3.7518
South Africa	rand	R	ZAR	0.0862	11.5995
United Arab Emirates	dirham	—	AED	0.2723	3.673

Note: SDR from the International Monetary Fund; based on exchange rates for U.S., British, and Japanese currencies. Quotes based on trading among banks of \$1 million and more, as quoted at 4 P.M. ET by Reuters. Rates are drawn from The Wall Street Journal online on February 19, 2015.

Exhibit 5.8 Exchange Rates: New York Closing Snapshot

Quoted rates

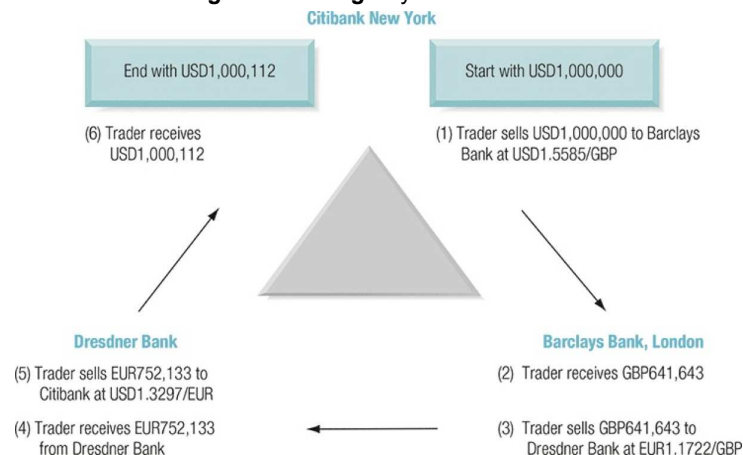
Citibank quotes U.S. dollars per euro	USD1.3297 = 1 EUR
Barclays Bank quotes U.S. dollars per pound sterling	USD1.5585 = 1 GBP
Dresdner Bank quotes euros per pound sterling	EUR1.1722 = 1 GBP

Cross rate calculation based on Citibank and Barclays Bank quotes

$$\frac{\text{USD1.5585/GDP}}{\text{USD1.3297/EUR}} = \text{EUR1.721/GDP}$$

Which is .001 less than the Dressner Bank quote, which results in triangular arbitrage

Exhibit 5.10 *Triangular Arbitrage* by a Market Trader



Percentage Change in Spot Rates

Foreign Currency Terms: Foreign currency price (price P_s) of the home currency (the unit, \$) is used
→ Measuring a change in the spot rate for quotations expressed in home currency terms (**direct quotations**)

$$\% \Delta = \frac{\text{Beginning Rate} - \text{Ending Rate}}{\text{Ending Rate}} \times 100$$

Home Currency Terms: Home currency price (the price) for a foreign currency (the unit) is used
→ Quotations expressed in foreign currency terms (**indirect quotations**)

$$\% \Delta = \frac{\text{Ending Rate} - \text{Beginning Rate}}{\text{Beginning Rate}} \times 100$$

Forward Quotations

Spot rates are typically quoted on an outright basis (meaning all digits expressed)
Forward rates are typically quoted in points or pips (the last digits of a currency quotation).
- Forward rates of one year or less maturity are termed cash rates; for longer than one-year they are called swap rates.

Exhibit 5.11: the bid & ask spot quotes are outright quotes, but the forwards are stated as points from the spot rate.

Term	Euro: Spot and Forward (\$/€)				Japanese Yen: Spot and Forward (¥/\$)			
	Bid		Ask		Bid		Ask	
	Points	Rate	Points	Rate	Points	Rate	Points	Rate
Cash rates								
Spot		1.0897		1.0901		118.27		118.37
1 week	3	1.0900	4	1.0905	-10	118.17	-9	118.28
1 month	17	1.0914	19	1.0920	-51	117.76	-50	117.87
2 months	35	1.0932	36	1.0937	-95	117.32	-93	117.44
3 months	53	1.0950	54	1.0955	-143	116.84	-140	116.97
4 months	72	1.0969	76	1.0977	-195	116.32	-190	116.47
5 months	90	1.0987	95	1.0996	-240	115.87	-237	116.00
6 months	112	1.1009	113	1.1014	-288	115.39	-287	115.50
9 months	175	1.1072	177	1.1078	-435	113.92	-429	114.08
1 year	242	1.1139	245	1.1146	-584	112.43	-581	112.56
Swap rates								
2 years	481	1.1378	522	1.1423	-1150	106.77	-1129	107.08
3 years	750	1.1647	810	1.1711	-1748	100.79	-1698	101.39
4 years	960	1.1857	1039	1.1940	-2185	96.42	-2115	97.22
5 years	1129	1.2026	1276	1.2177	-2592	92.35	-2490	93.47

Forward Quotations in Percentage Terms

The percent per annum deviation of the forward from the spot rate is termed the forward premium.

As with the calculation of percentage changes in spot rates, the forward premium - which may be either a positive (a premium) or negative value (a discount) depends upon the designated home (or base) currency.

step 1: which currency is unit currency

step 2: employ traditional (unit currency) / nontraditional formula (not unit currency)

- spot rate: 23,000 VND/ USD

- FW rate: 24,000 VND/ USD

→ calculate fw premium/ discount on dollar

ex:

step 1: dollar: unit currency

step 2:

% change on usd $F = (F - S) / S \times 100\% \times 360/m$

% change on vnd → employ nontraditional formula: $S-F/F \times 100\% \times 360/m$

International Parity Conditions

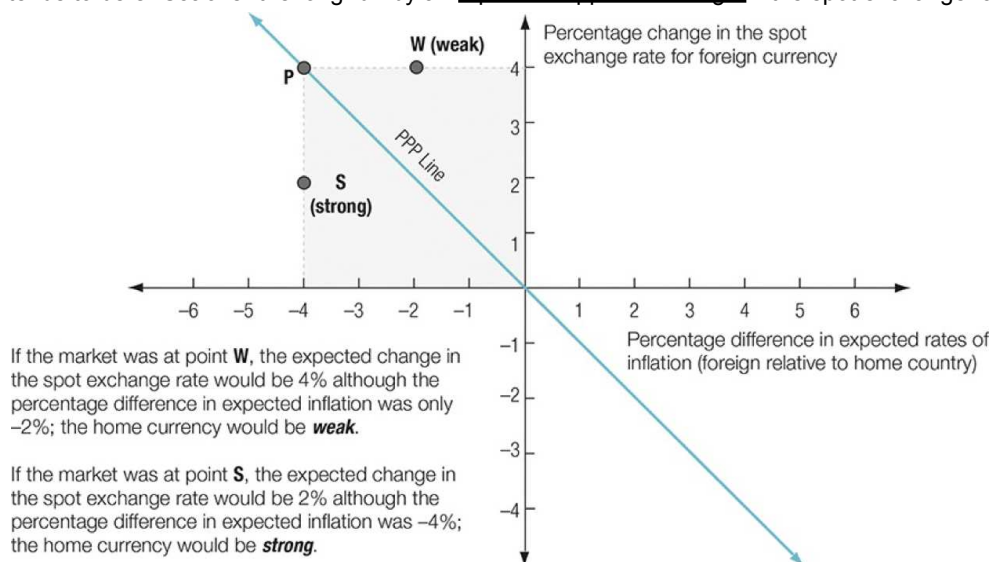
International Parity Conditions

- Some fundamental questions managers of MNEs, international portfolio investors, importers, exporters and government officials must deal with every day are:
 - What are the determinants of exchange rates?
 - Are changes in exchange rates predictable?
- The economic theories that link exchange rates, price levels, and interest rates together are called international parity conditions.
- These international parity conditions form the core of the financial theory that is unique to international finance.
- These theories do not always work out to be “true” when compared to what students and practitioners observe in the real world, but they are central to any understanding of how multinational business is conducted and funded in the world today.
- The mistake is often not with the theory itself, but with the interpretation and application of said theories

Prices and Exchange Rates

- LAW OF PRICE:** If the identical product or service can be:
 - sold in two different markets; and
 - no restrictions exist on the sale; and
 - transportation costs of moving the product between markets are equal, then
 - the product's price should be the same in both markets.
- A primary principle of competitive markets is that prices will equalize across markets if frictions (transportation costs) do not exist.
- Comparing prices then, would require only a conversion from one currency to the other:
 - $P\$ \times S = P¥$
 - Where the product price in U.S. dollars is ($P\$$), the spot exchange rate is (S) and the price in Yen is ($P¥$).
- If the law of one price were true for all goods and services, the purchasing power parity (PPP) exchange rate could be found from any individual set of prices.
- By comparing the prices of identical products denominated in different currencies, we could determine the “real” or PPP exchange rate that should exist if markets were efficient.
- This is the absolute version of the PPP theory.
- If the assumptions of the absolute version of the PPP theory are relaxed a bit more, we observe what is termed relative purchasing power parity (RPPP).
- RPPP holds that PPP is not particularly helpful in determining what the spot rate is today, but that the relative change in prices between two countries over a period of time determines the change in the exchange rate over that period.
- More specifically, with regard to RPPP:

“If the spot exchange rate between two countries starts in equilibrium, any change in the differential rate of inflation between them tends to be **offset** over the long run by an equal but opposite change in the spot exchange rate.”



- Empirical testing of PPP and the law of one price has been done, but has not proved PPP to be accurate in predicting future exchange rates.
- Two general conclusions can be made from these tests:
 - PPP holds up well over the very long run but poorly for shorter time periods; and,
 - the theory holds better for countries with relatively high rates of inflation and underdeveloped capital markets.
- Individual national currencies often need to be evaluated against other currency values to determine relative purchasing power.
- Objective: discover whether a nation's exchange rate is “overvalued” or “undervalued” in terms of PPP → nominal effective exchange rate
- This problem is often dealt with through the calculation of exchange rate indices such as the nominal effective exchange rate index.

Exchange Rate Pass-Through

- Is a measure of the response of imported and exported product prices to changes in exchange rates.

Prices and Exchange Rates

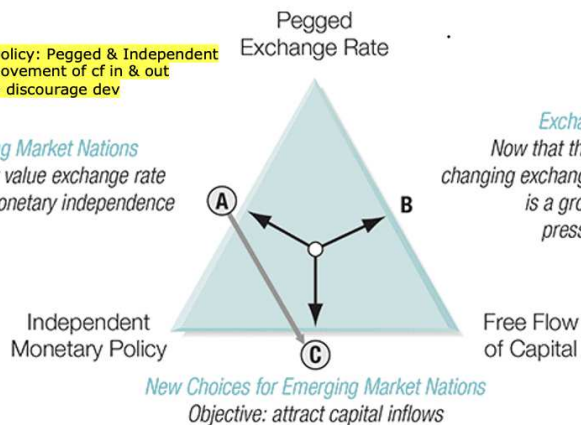
- Price elasticity of demand is an important factor when determining pass-through levels.
- The own-price elasticity of demand for any good is the percentage change in quantity of the good demanded as a result of the percentage change in the good's own price.

Pass-Through, Impossible Trinity, and Emerging Markets

Only VN: old policy: Pegged & Independent

- Not allow movement of cf in & out
- - eco dev → discourage dev
- + eco sta

Emerging Market Nations
traditionally value exchange rate stability & monetary independence



Many emerging market countries have chosen to move from Point A to Point C, exchanging fixed exchange rates for the chance of attracting capital inflows. The result is that these countries are now the subject to varying levels of exchange rate pass-through.

Interest Rates and Exchange Rates

- The **Fisher effect** states that nominal interest rates in each country are equal to the required real rate of return plus compensation for expected inflation.
This equation reduces to (in approximate form) →
- Empirical tests (using ex-post) national inflation rates have shown the Fisher effect usually exists for short-maturity government securities (treasury bills and notes).

$$i = r + \pi$$

- i = nominal interest rate
- r = real interest rate and
- π = expected inflation.

- The relationship between the percentage change in the spot exchange rate over time and the differential between comparable interest rates in different national capital markets is known as the international Fisher effect.
- "Fisher-open,"** as it is termed, states that the spot exchange rate should change in an equal amount but in the opposite direction to the difference in interest rates between two countries.
More formally →
- Justification for the international Fisher effect is that investors must be rewarded or penalized to offset the expected change in exchange rates.

$$\frac{S_1 - S_2}{S_2} \times 100 = i^S - i^Y$$

i^S, i^Y : respective national interest rates
 S : spot exchange rate using indirect quotes ($\text{¥}/\text{\$}$).

The Forward Rate

- A forward rate is an exchange rate quoted for settlement at some **future date**.
- A forward exchange agreement between currencies states the rate of exchange at which a foreign currency will be bought forward or sold forward at a specific date in the future.
- The forward rate is calculated for any specific maturity by adjusting the current spot exchange rate by the ratio of eurocurrency interest rates of the same maturity for the two subject currencies.
 - Ex: the 90-day forward rate for the Swiss franc/U.S. dollar exchange rate ($F_{90}^{\text{SF}/\$}$) is found by multiplying the current spot rate ($S^{\text{SF}/\$}$) by the ratio of the 90-day euro-Swiss franc deposit rate (i^{SF}) over the 90-day eurodollar deposit rate ($i^{\$}$).

Formulaic representation of the forward rate:

$$F_{90}^{\text{SF}/\$} = S^{\text{SF}/\$} \times \frac{1 + \left(i^{\text{SF}} \times \frac{90}{360} \right)}{1 + \left(i^{\$} \times \frac{90}{360} \right)}$$

The forward premium or discount is the percentage difference between the spot and forward exchange rate, stated in annual percentage terms.

$$f^{\text{SF}} = \frac{\text{Spot} - \text{Forward}}{\text{Forward}} \times \frac{360}{\text{Days}} \times 100.$$

This is the case when the foreign currency price of the home currency is used ($\text{SF}/\text{\$}$).

Interest Rate Parity (IRP)

- The theory of IRP provides the linkage between the foreign exchange markets and the international money markets.
- The theory states: The difference in the national interest rates for securities of similar risk and maturity should be equal to, but opposite in sign to, the forward rate discount or premium for the foreign currency, except for transaction costs.

See Exhibit 6.6: Interest Rate Parity (IRP)

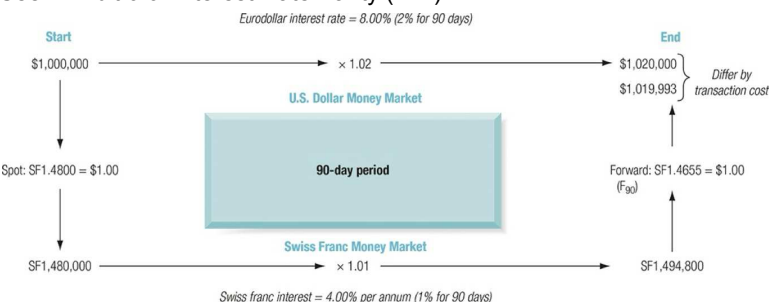
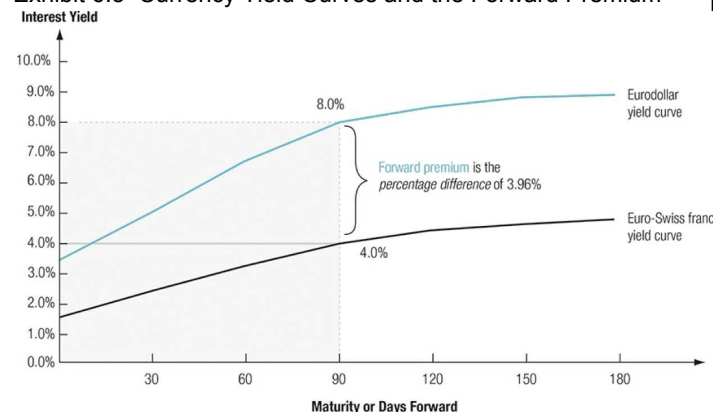


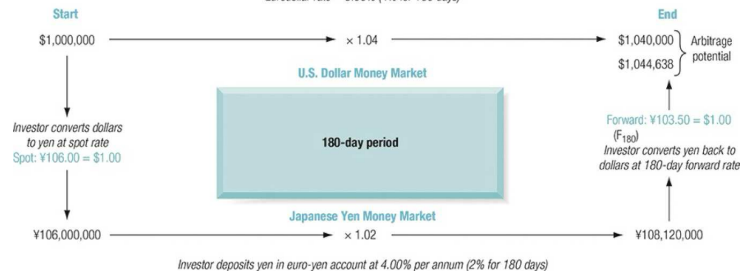
Exhibit 6.5 Currency Yield Curves and the Forward Premium



Covered Interest Arbitrage (CIA) buôn chứng khoán

- The spot and forward exchange rates are not, however, constantly in the state of equilibrium described by interest rate parity.
- When the market is not in equilibrium, the potential for “risk-less” or arbitrage profit exists.
- The arbitrageur will exploit the imbalance by investing in whichever currency offers the higher return on a covered basis.

Covered Interest Arbitrage (CIA) kinh doanh chênh lệch giá bảo đảm lãi suất

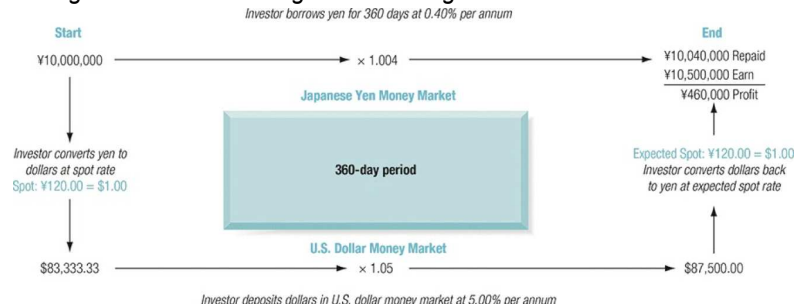


Uncovered Interest Arbitrage (UIA)

- In the case of uncovered interest arbitrage (UIA), investors borrow in countries and currencies exhibiting relatively low interest rates and convert the proceeds into currencies that offer much higher interest rates (6.8%).
- The transaction is “uncovered” because the investor does not sell the higher yielding currency proceeds forward, choosing to remain uncovered and accept the currency risk of exchanging the higher yield currency into the lower yielding currency at the end of the period.
- U do not enter forward contract → gain/loss depends on your luck → gambling

Uncovered Interest Arbitrage (UIA): The Yen Carry Trade

→ higher risk but much higher return/huger loss



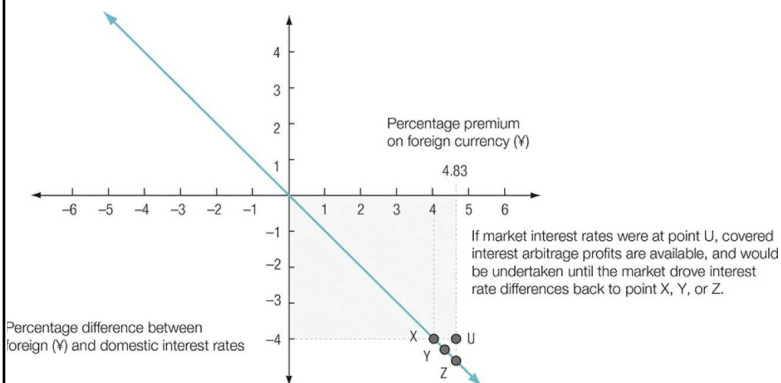
In the yen carry trade, the investor borrows Japanese yen at relatively low interest rates, converts the proceeds to another currency such as the U.S. dollar where the funds are invested at a higher interest rate for a term. At the end of the period, the investor exchanges the dollars back to yen to repay the loan, pocketing the difference as arbitrage profit. If the spot rate at the end of the period is roughly the same as at the start, or the yen has fallen in value against the dollar, the investor profits. If, however, the yen were to appreciate versus the dollar over the period, the investment may result in significant loss.

Equilibrium between Interest Rates and Exchange Rates

The conditions necessary for equilibrium between interest rates and exchange rates. →

The disequilibrium situation, denoted by point U, is located off the interest rate parity line.

However, the situation represented by point U is unstable because all investors have an incentive to execute the same covered interest arbitrage, which is virtually risk-free.

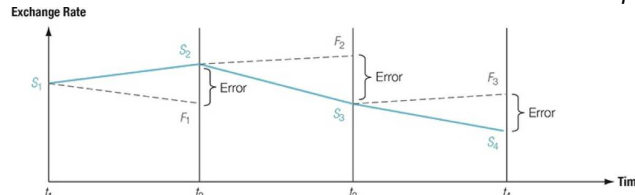


Some forecasters believe that forward exchange rates are unbiased predictors of future spot exchange rates.

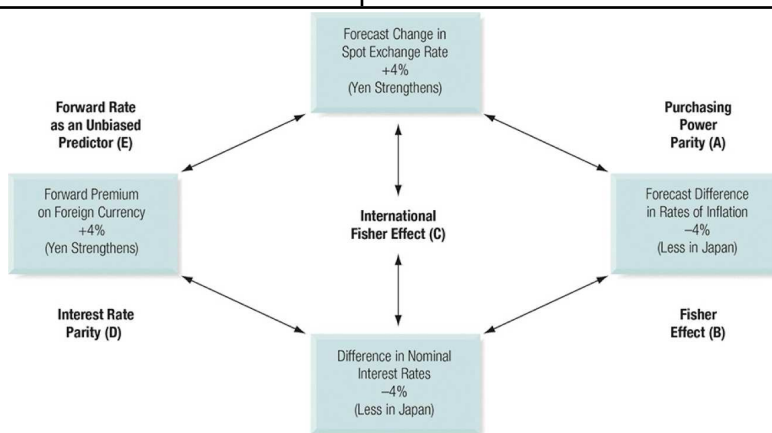
Intuitively this means that the distribution of possible actual spot rates in the future is centered on the forward rate.

Unbiased prediction simply means that the forward rate will, on average, overestimate and underestimate the actual future spot rate in equal frequency and degree.

Forward Rate as an Unbiased Predictor of Future Spot



The forward rate available “today” (t) for delivery at a future time ($t + 1$) is used as a forecast or predictor of the spot rate at time $t + 1$. The difference between the spot rate which then occurs and the forward rate is the forecast error. When the forward rate is termed an “unbiased predictor of the future spot rate” it means that the errors are normally distributed around the mean future spot rate (the sum of the errors equal zero).



Foreign Currency Derivatives: Futures and Options

Foreign Currency Derivatives and Swaps

- Financial management of the MNE in the 21st century involves financial derivatives.
- These derivatives, so named because their values are derived from underlying assets, are a powerful tool used in business today.
- These instruments can be used for two very distinct management objectives:
 - Speculation: use of derivative instruments to take a position in the expectation of a profit
 - Hedging: use of derivative instruments to reduce the risks associated with the everyday management of corporate cash flow
- Derivatives** are used by firms to achieve one or more of the following individual benefits:
 - Permit firms to achieve payoffs that they would not be able to achieve without derivatives, or could achieve only at greater cost
 - Hedge risks that otherwise would not be possible to hedge
 - Make underlying markets more efficient
 - Reduce volatility of stock returns
 - Minimize earnings volatility
 - Reduce tax liabilities
 - Motivate management (agency theory effect)

Foreign Currency Futures

- A foreign currency futures contract is an alternative to a forward contract that calls for future delivery of a standard amount of foreign exchange at a fixed time, place and price.
- It is similar to futures contracts that exist for commodities such as cattle, lumber, interest-bearing deposits, gold, etc.
- In the U.S., the most important market for foreign currency futures is the International Monetary Market (IMM), a division of the Chicago Mercantile Exchange.
- Contract specifications are established by the exchange on which futures are traded.
- Major features that are standardized are:
 - Contract size
 - Method of stating exchange rates
 - Maturity date
 - Last trading day
 - Collateral and maintenance margins
 - Settlement
 - Commissions
 - Use of a clearinghouse as a counterparty
- Exhibit 7.1 provides a description of futures contracts for the Mexican peso

Maturity	Open	High	Low	Settle	Change	Lifetime		Open Interest
						High	Low	
Mar	0.10953	0.10988	0.10930	0.10958	...	0.11000	0.09770	34,481.00
June	0.10790	0.10795	0.10778	0.10773	...	0.10800	0.09730	3,405.00
Sept	0.10615	0.10615	0.10610	0.10573	...	0.10615	0.09930	1,481.00

All contracts are for 500,000 Mexican pesos. "Open" means the opening price on the day. "High" means the high price on the day. "Low" indicates the lowest price on the day. "Settle" is the closing price on the day. "Change" indicates the change in the settle price from the previous day's close. "High" and "Low" to the right of "Change" indicate the highest and lowest prices this specific contract (as defined by its maturity) has experienced over its trading history. "Open Interest" indicates the number of contracts outstanding.

Foreign Currency Futures

- Foreign currency futures contracts differ from forward contracts in a number of important ways:
 - Futures are standardized in terms of size while forwards can be customized
 - Futures have fixed maturities while forwards can have any maturity (both typically have maturities of one year or less)
 - Trading on futures occurs on organized exchanges while forwards are traded between individuals and banks
 - Futures have an initial margin that is market to market on a daily basis while only a bank relationship is needed for a forward
 - Futures are rarely delivered upon (settled) while forwards are normally delivered upon (settled)

Foreign Currency Options

- A foreign currency option is a contract giving the option purchaser (the buyer) the right, but not the obligation, to buy or sell a given amount of foreign exchange at a fixed price per unit for a specified time period (until the maturity date).
- There are two basic types of options, puts and calls.
 - A call is an option to buy foreign currency
 - A put is an option to sell foreign currency
- The buyer of an option is termed the holder, while the seller of the option is referred to as the writer or grantor.
- Every option has three different price elements:
 - The exercise or strike price: the exchange rate at which the foreign currency can be purchased (call) or sold (put)
 - The premium: the cost, price, value of the option
 - The underlying or actual spot exchange rate in the market
- American option gives the buyer the right to exercise the option at any time between the date of writing and the expiration or maturity.
- A European option can be exercised only on its expiration date, not before.
- The premium, or option price, is the cost of the option.
- An option whose exercise price is the same as the spot price of the underlying currency is said to be at-the-money (ATM).
- An option that would be profitable, excluding the cost of the premium, if exercised immediately is said to be in-the-money (ITM).

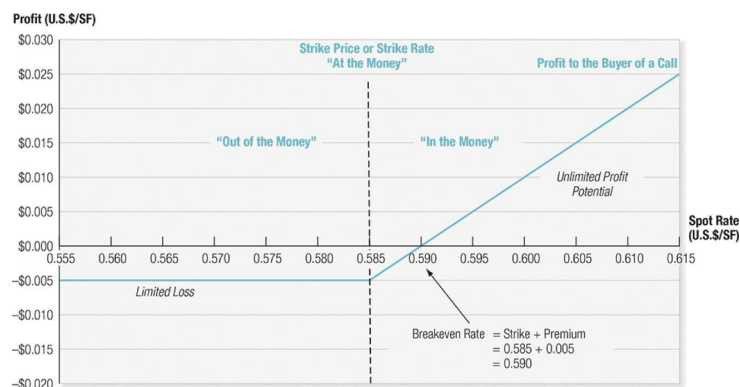
- An option that would not be profitable, again excluding the cost of the premium, if exercised immediately is referred to as out-of-the money (OTM).
- In the past three decades, the use of foreign currency options as a hedging tool and for speculative purposes has blossomed into a major foreign exchange activity.
- Options on the over-the-counter (OTC) market can be tailored to the specific needs of the firm but can expose the firm to counterparty risk.
- Options on organized exchanges are standardized, but counterparty risk is substantially reduced.
- Exhibit 7.2 shows a published quote for the Swiss Franc.

Option and Underlying	Strike Price	Calls – Last			Puts – Last		
		Aug	Sep	Dec	Aug	Sep	Dec
58.51	56.0	—	—	2.76	0.04	0.22	1.16
58.51	56.5	—	—	—	0.06	0.30	—
58.51	57.0	1.13	—	1.74	0.10	0.38	1.27
58.51	57.5	0.75	—	—	0.17	0.55	—
58.51	58.0	0.71	1.05	1.28	0.27	0.89	1.81
58.51	58.5	0.50	—	—	0.50	0.99	—
58.51	59.0	0.30	0.66	1.21	0.90	1.36	—
58.51	59.5	0.15	0.40	—	2.32	—	—
58.51	60.0	—	0.31	—	2.32	2.62	3.30

Each option = 62,500 Swiss francs. The August, September, and December listings are the option maturities or expiration dates.

Buyer of a Call Option

- Buyer of an option only exercises his/her rights if the option is profitable.
- In the case of a call option, as the spot price of the underlying currency moves up, the holder has the possibility of unlimited profit.
- Exhibit 7.3 shows a static profit and loss diagram for the purchase of a Swiss Franc Call Option. Notice how the purchaser makes a profit as the franc appreciates vs. the dollar because the purchaser has the right to purchase the franc at a pre-specified/lower price than the current spot price.



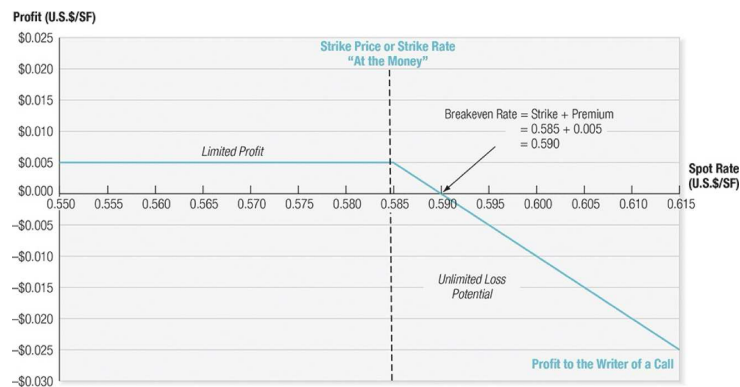
The **buyer of a call option** on has unlimited profit potential (*in the money*), and limited loss potential, the amount of the premium (*out of the money*).

Exhibit 7.3 Profit and Loss for the Buyer of a Call Option

Option Market Speculation

Writer of a call (see Exhibit 7.4):

- What the holder, or buyer of an option loses, the writer gains
- The maximum profit that the writer of the call option can make is limited to the premium
- If the writer wrote the option naked, that is without owning the currency, the writer would now have to buy the currency at the spot and take the loss delivering at the strike price
- The amount of such a loss is unlimited and increases as the underlying currency rises
- Even if the writer already owns the currency, the writer will experience an opportunity loss

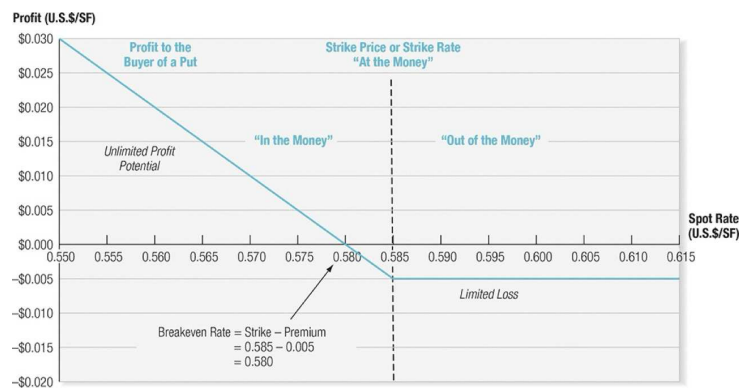


The **writer of a call option** on has unlimited loss potential and limited profit potential, the amount of the premium.

Exhibit 7.4 Profit and Loss for the Writer of a Call Option

Buyer of a Put (see Exhibit 7.5):

- The basic terms of this example are similar to those just illustrated with the call
- The buyer of a put option, however, wants to be able to sell the underlying currency at the exercise price when the market price of that currency drops (not rises as in the case of the call option)
- If the spot price drops to \$0.575/SF, the buyer of the put will deliver francs to the writer and receive \$0.585/SF
- At any exchange rate above the strike price of 58.5, the buyer of the put would not exercise the option, and would lose only the \$0.05/SF premium
- The buyer of a put (like the buyer of the call) can never lose more than the premium paid up front

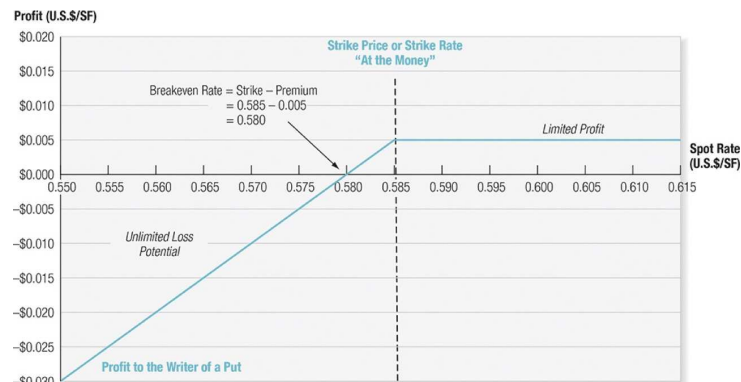


The **buyer of a put option** on has unlimited profit potential (*in the money*), and limited loss potential, the amount of the premium (*out of the money*).

Exhibit 7.5 Profit and Loss for the Buyer of a Put Option

Seller (writer) of a put (see Exhibit 7.6):

- In this case, if the spot price of francs drops below 58.5 cents per franc, the option will be exercised
- Below a price of 58.5 cents per franc, the writer will lose more than the premium received from writing the option (falling below break-even)
- If the spot price is above \$0.585/SF, the option will not be exercised and the option writer will pocket the entire premium



The writer of a put option has limited profit potential, the premium, and an unlimited loss potential.

Exhibit 7.6 Profit and Loss for the Writer of a Put Option

Option Pricing and Valuation

- The pricing of any currency option combines six elements:
 - Present spot rate
 - Time to maturity
 - Forward rate for matching maturity
 - U.S. dollar interest rate
 - Foreign currency interest rate
 - Volatility (standard deviation of daily spot price movements)
- The total value (premium) of an option is equal to the intrinsic value plus time value.
- Intrinsic value is the financial gain if the option is exercised immediately.
 - For a call option, intrinsic value is zero when the strike price is above the market price
 - When the spot price rises above the strike price, the intrinsic value become positive
 - Put options behave in the opposite manner
 - On the date of maturity, an option will have a value equal to its intrinsic value (zero time remaining means zero time value)
- The time value of an option exists because the price of the underlying currency, the spot rate, can potentially move further into the money between the present time and the option's expiration date.

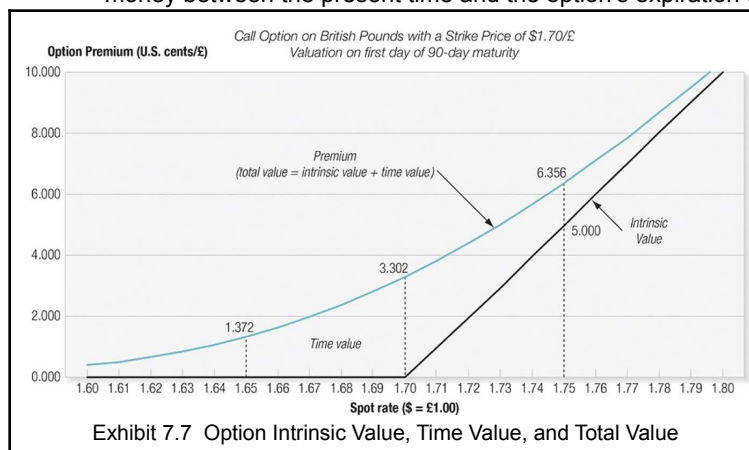


Exhibit 7.7 Option Intrinsic Value, Time Value, and Total Value

Strike Rate (\$/£)	Spot Rate (\$/£)	Money	Call Premium (U.S. cents/£)	=	Intrinsic Value (U.S. cents/£)	+	Time Value (U.S. cents/£)	Option Delta (0 to 1)
1.70	1.75	In-the-money (ITM)	6.37	=	5.00	+	1.37	0.71
1.70	1.70	At-the-money (ATM)	3.30	=	0.00	+	3.30	0.50
1.70	1.65	Out-of-the-money (OTM)	1.37	=	0.00	+	1.37	0.28

Exhibit 7.8 Call Option Premiums: Intrinsic Value and Time Value Components

Currency Option Pricing Sensitivity

- If currency options are to be used effectively, either for the purposes of speculation or risk management, the individual trader needs to know how option valuations (premiums) react to their various components.
- 6 sensitivities:
 - The impact of changing forward rates
 - The impact of changing spot rates
 - The impact of time to maturity
 - The impact of changing volatility
 - The impact of changing interest differentials
 - The impact of alternative option strike prices

Forward Rate Sensitivity

- Standard foreign currency options are priced around the forward rate because the current spot rate and both the domestic and foreign interest rates (home currency and foreign currency rates) are included in the option premium calculation.
- The forward rate is central to valuation.
- The option-pricing formula calculates a subjective probability distribution centered on the forward rate.

Spot Rate Sensitivity (delta)

- If the current spot rate falls on the side of the option's strike price—which would induce the option holder to exercise the option upon expiration - the option also has an intrinsic value.
- The sensitivity of the option premium to a small change in the spot exchange rate is called the delta.
- Delta varies between +1 and 0 for a call option and -1 and 0 for a put option.
- As an option moves further in-the-money, delta rises toward 1.0. As an option moves further out-of-the-money, delta falls toward zero.
- Rule of Thumb: The higher the delta (deltas of > .7 are high) the greater the probability of the option expiring in-the-money.

Time to Maturity: Value and Deterioration (theta)

- Option values increase with the length of time to maturity. The expected change in the option premium from a small change in the time to expiration is termed theta.
- Theta is calculated as the change in the option premium over the change in time. Theta is based not on a linear relationship with time, but rather the square root of time.
- Option premiums deteriorate at an increasing rate as they approach expiration.
- Rule of Thumb:** A trader will normally find longer-maturity options better values, giving the trader the ability to alter an option position without suffering significant time value deterioration.

Sensitivity to Volatility (lambda)

- Option volatility is the standard deviation of daily percentage changes in the underlying exchange rate.
- The primary problem with volatility is that there is no single method for its calculation. Volatility is viewed three ways:
 - historic, where the volatility is drawn from a recent period of time;
 - forward-looking, where the historic volatility is altered to reflect expectations about the future period over which the option will exist; and
 - implied, where the volatility is backed out of the market price of the option. Selected implied volatilities for a number of currency pairs are listed in Exhibit 7.9.
- Rule of Thumb:** Traders who believe volatilities will fall significantly in the near-term will sell (write) options now, hoping to buy them back for a profit immediately after volatilities fall causing option premiums to fall.

Currency (cross)	Symbol	1 week	1 month	2 month	3 month	6 month	1 year	2 year	3 year
European euro	EUR	8.1	7.4	7.4	7.4	7.8	8.5	9.0	9.3
Japanese yen	JPY	12.3	11.4	11.1	11.0	11.0	11.2	11.8	12.7
Swiss franc	CHF	8.9	8.4	8.4	8.4	8.9	9.5	9.8	9.9
British pound	GBP	7.7	7.3	7.2	7.1	7.3	7.5	7.9	8.2
Canadian dollar	CAD	6.4	6.4	6.3	6.4	6.7	7.1	7.4	7.6
Australian dollar	AUD	11.2	10.7	10.5	10.3	10.4	10.6	10.8	11.0
British pound/euro	GBPEUR	6.7	6.4	6.5	6.4	6.8	7.3	7.6	7.8
Euro/Japanese yen	EURJPY	11.6	11.1	11.2	11.3	11.8	12.6	13.4	14.1

Source: Federal Reserve Bank of New York.

Note: These implied volatility rates are averages of mid-level rates for bid and ask "at-money quotations" on selected currencies at 11 A.M. on the last business day of the month, September 30, 2013.

Exhibit 7.9 Foreign Currency Implied Volatilities (percent)

Sensitivity to Changing Interest Rate Differentials (rho and phi)

- The expected change in the option premium from a small change in the domestic interest rate (home currency) is termed rho.
- The expected change in the option premium from a small change in the foreign interest rate (foreign currency) is termed phi.
- Rule of Thumb:** A trader who is purchasing a call option on foreign currency should do so before the domestic interest rate rises. This will allow the trader to purchase the option before its price increases.

Alternative Strike Prices and Option Premiums

- A firm purchasing an option in the over-the-counter market may choose its own strike rate.
- Options with strike rates that are already in-the-money will have both intrinsic and time value elements.
- Options with strike rates that are out-of-the-money will have only a time value component.

Greek	Definition	Interpretation
<i>Delta</i>	Expected change in the option premium for a small change in the spot rate	The higher the delta, the more likely the option will move in-the-money
<i>Theta</i>	Expected change in the option premium for a small change in time to expiration	Premiums are relatively insensitive until the final 30 or so days
<i>Lambda</i>	Expected change in the option premium for a small change in volatility	Premiums rise with increases in volatility
<i>Rho</i>	Expected change in the option premium for a small change in the domestic interest rate	Increases in domestic interest rates cause increasing call option premiums
<i>Phi</i>	Expected change in the option premium for a small change in the foreign interest rate	Increases in foreign interest rates cause decreasing call option premiums

Summary of Option Premium Components